

2

Current Electricity

- ★ Current (i) $\Rightarrow$ Rate of flow of charge, scalar quantity

$$i = \frac{dq}{dt}$$

- ★ Inst Current = $i = \frac{dq}{dt}$

- ★ Average Current = $\langle i \rangle = \frac{\int i \cdot dt}{\Delta t} = \frac{\Delta q}{\Delta t}$

- ★ $\langle \vec{V} \rangle =$ Average velocity = $\frac{\int v dt}{\int dt}$ $\langle \text{Kaddu} \rangle = \frac{\int \text{Kaddu} dt}{\int dt}$

Q. Given $i = 3t^2$

(a) Find current at $t = 2$ sec

Sol. $i = 3 \times 2^2 = 12$

(b) Finds average current from $t = 0 \rightarrow t = 2$

$$\langle i \rangle = \frac{\int_0^2 i dt}{\int_0^2 dt} = \frac{\int_0^2 3t^2 dt}{\int_0^2 dt} = \frac{8}{2} = 4$$

(c) $i = 3t^2$ Find the charge flow from $t = 0 \rightarrow t = 2$ sec

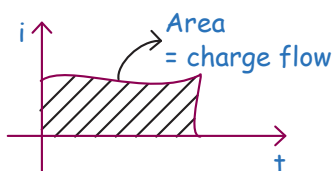
$$\Delta q = \int i dt = \int_0^2 3t^2 dt = 8$$

☆ $i = \frac{dq}{dt}$

$$\int dq = \int i dt$$

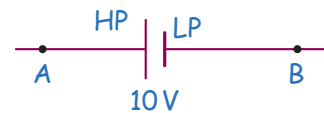
$$\Delta q = \int i dt = \text{Area}$$

↓
charge flow

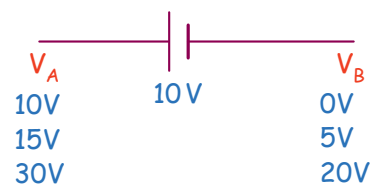


Abe suno! Current electricity mein sabse zaruri hotā hai circuit analysis, toh sequence change karke Saleem Bhaiya ke sequence mein padhte hain, from basic to advance."

- ☆ Ideal battery



$$V_A - V_B = 10$$



- ☆ Ohm's Law (derivation badh mein dekhege)

CE mein bohot jruri hai $V = iR$ lgana sikhna let's start it.

i $\Delta V = iR$

A B

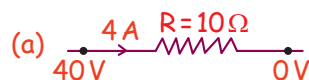
$V_A - V_B = iR$ or $V = iR$

pot. diff.

$i \rightarrow$ Resist mein H.P to L.P



- Q. Find current in the resistance in following cases.



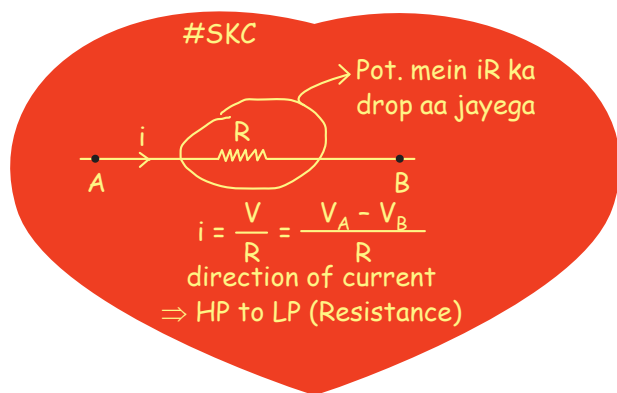
Sol. $i = \frac{40 - 0}{10} = 4$



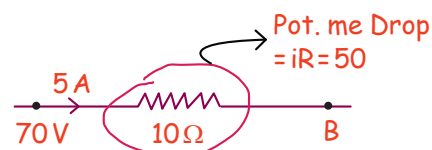
Sol. $i = \frac{50}{10} = 5$



Sol. $i = \frac{50 - (-10)}{10} = \frac{60}{10} = 6$



(d) Find V_B .

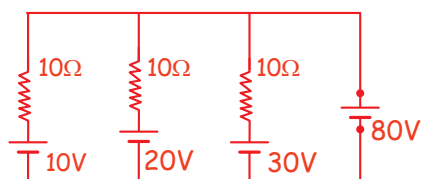


Sol. $V_B = 70 - 50 = 20V$

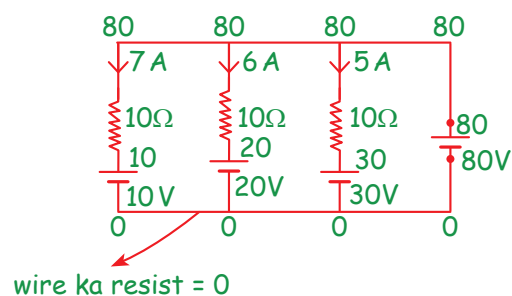
#SKC

Resistance mein
 current H.P to L.P flow
 krega but Battery mein
 kaise bi kar sakta
 hai

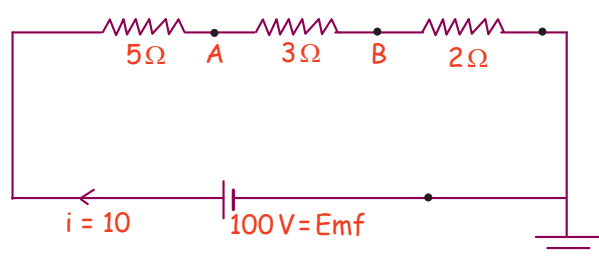
Q. Find current through each resistors.



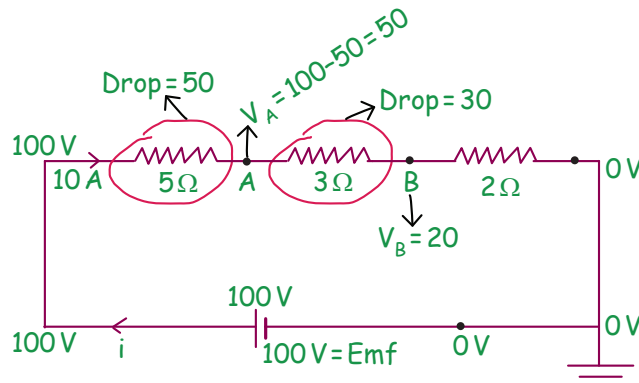
Sol.



Q. Find current and potential at A & B.



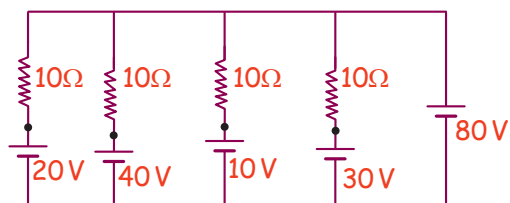
Sol.



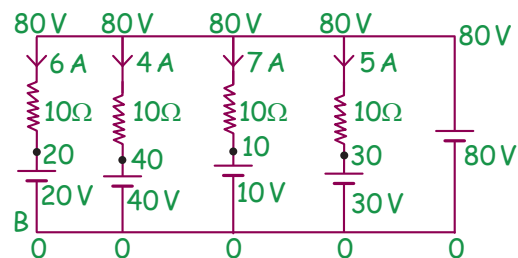
$i = \frac{E}{R_{eq}} = \frac{100}{10} = 10A$

$R_{eq} = 5 + 3 + 2 = 10\Omega$

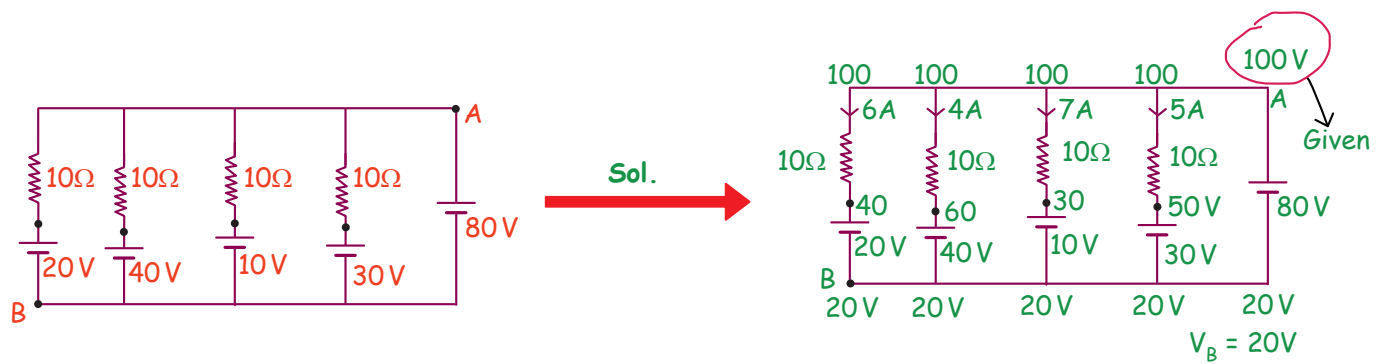
Q. Find current through each resistors ($R = 10\Omega$ each)



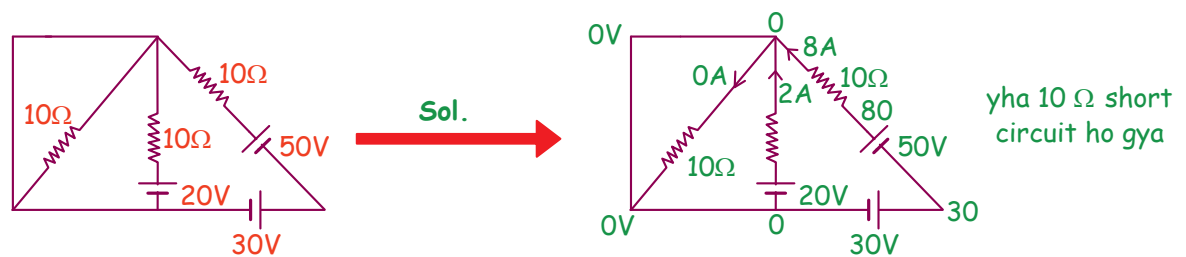
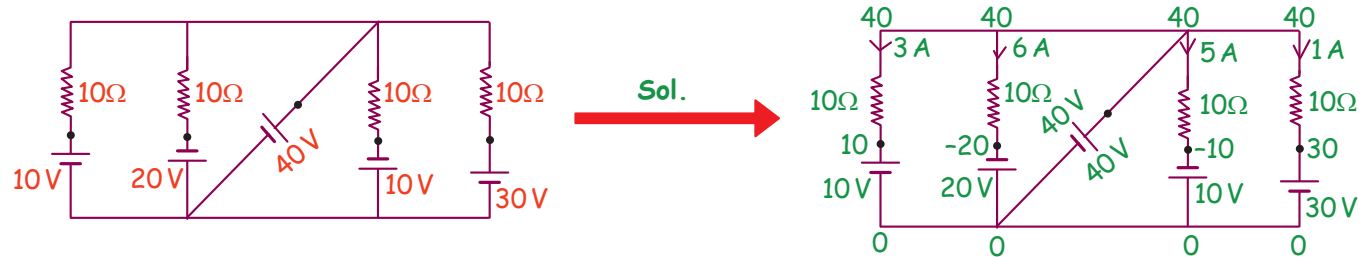
Sol.



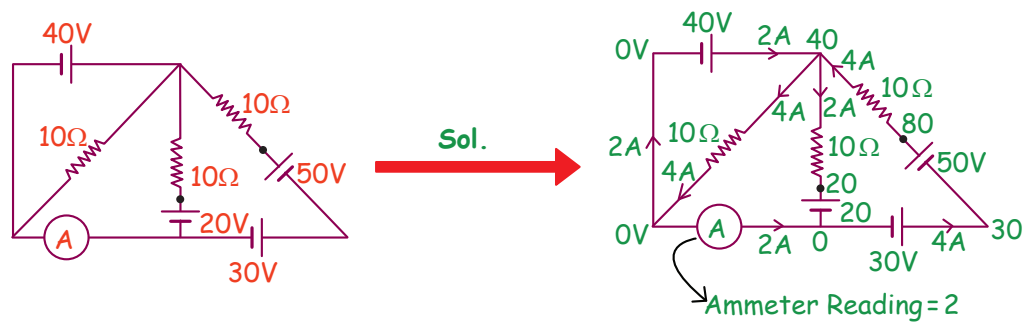
Q. If pot. at A is 100V, find V_B & Current. ($R = 10\Omega$ each)



Q. Find i in each resistance



Q. What is ammeter reading?



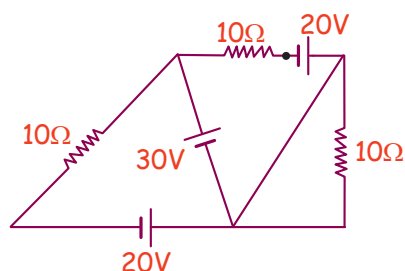


Maana ki no. of pages of book ko kam se kam rakhne ka pressure hai to minimise the MRP/- lekin circuit analysis ke sawaal hum bhar-bhar kar practice karenge. (Bcz it's vry imp)

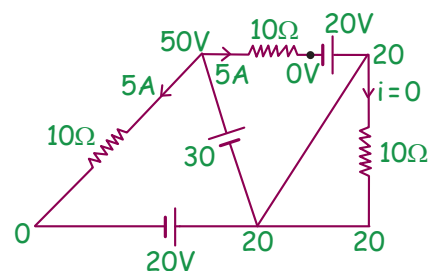


Abe ready ho na? Mujhe aisa lag raha hai ye tumhi ho..... Neeche ke left side ke saare sawaal khud se solve karo aur right side se match karao..... Aur jab saare sawaal ho jaaye, mujhe insta par confirmation do. (ID: saleem.nitt)

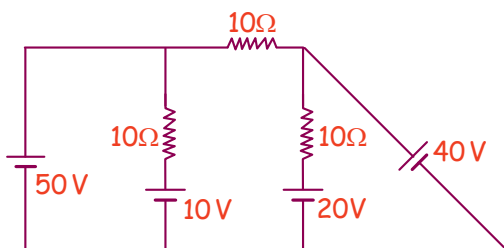
Q. Find current in each resistors.



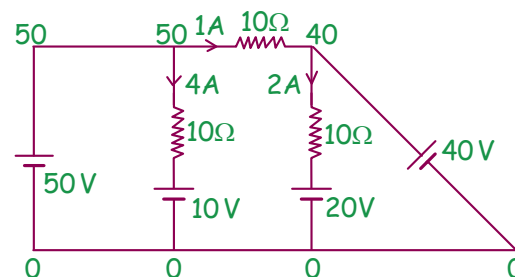
Sol.



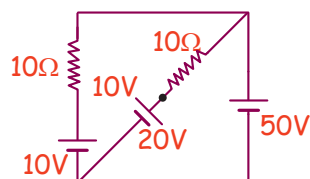
Q. Find current in each resistors.



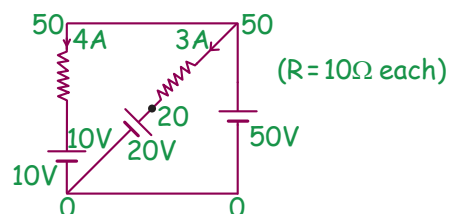
Sol.



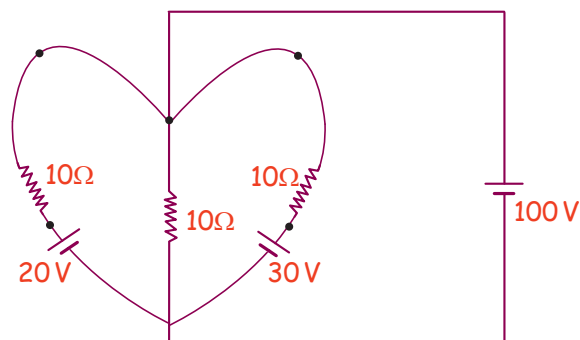
Q. Find current in each resistors.



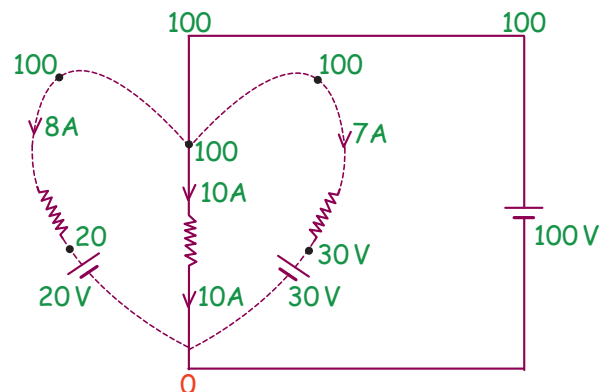
Sol.

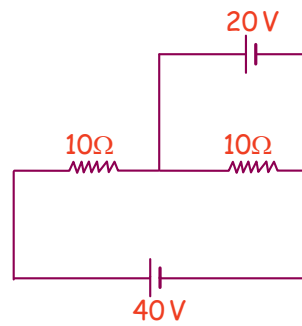


Q. Find current in each resistors.

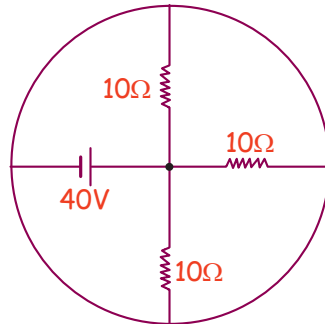
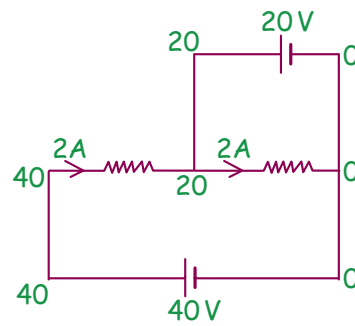


Sol.

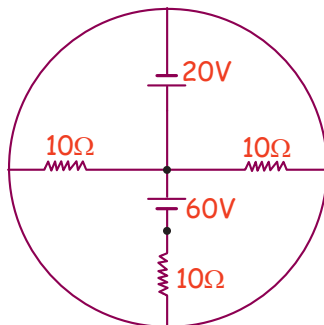
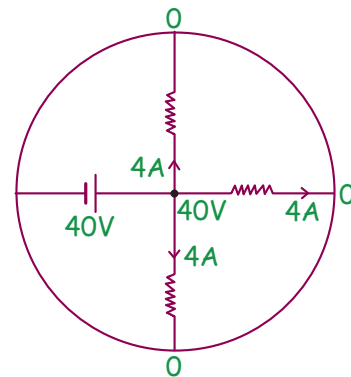




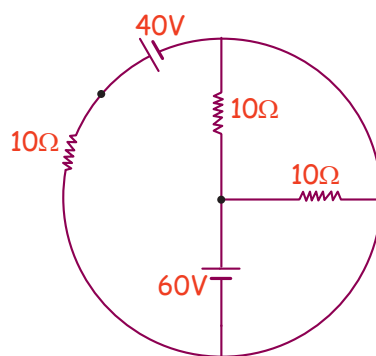
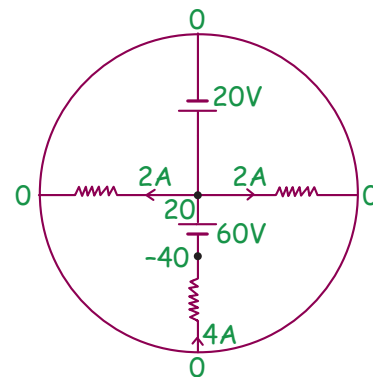
Sol.



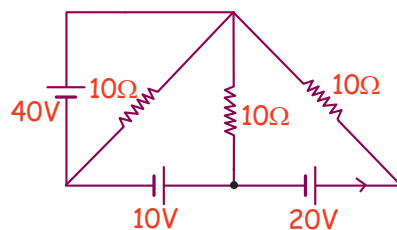
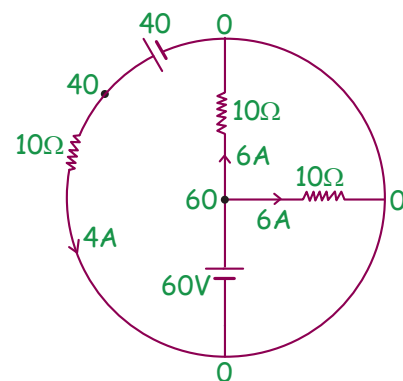
Sol.



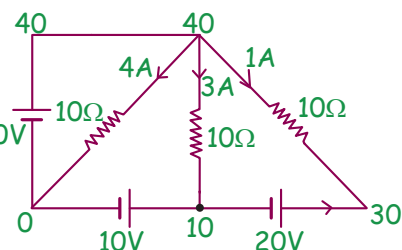
Sol.



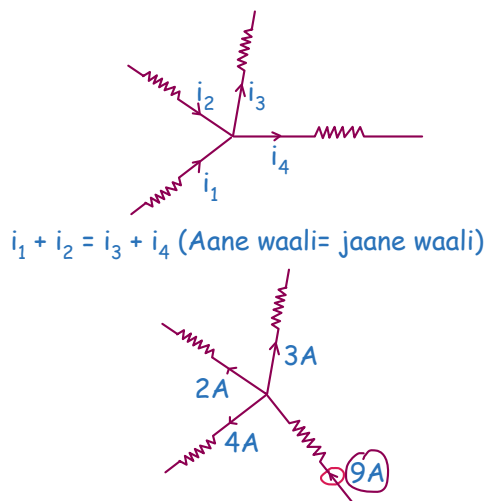
Sol.



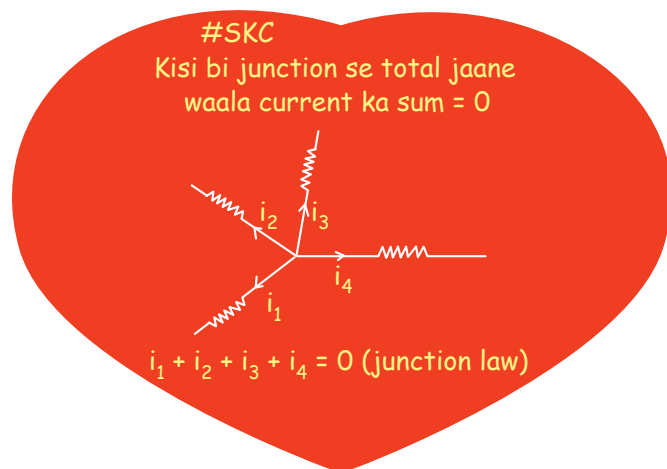
Sol.



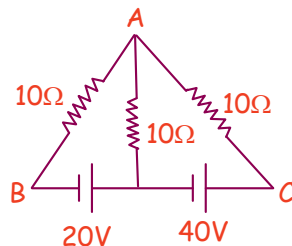
JUNCTION LAW



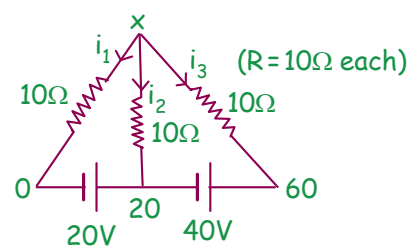
OR



Q. Find current in each resistance



Sol.



$$i_1 + i_2 + i_3 = 0$$

$$\frac{x - 0}{10} + \frac{x - 20}{10} + \frac{x - 60}{10} = 0$$

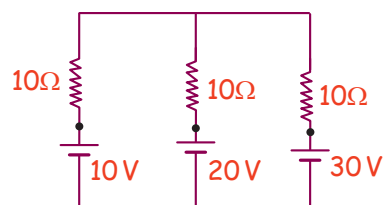
$$x = \frac{80}{3}$$

#SKC

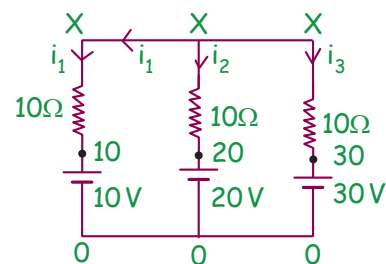
Agar B ka potential 0 maanu to C ka potential will be 60 volt lekin hum A ka potential nahi bata sakte, hum phans gaye..... Jahaan phans jao wahan EX ko yaad karo..... 😊

Abe emotional mat ho I mean jahaan phanse wahin potential X maan lo aur junction law lagaado

Q. Find current through each resistance.



Sol.



$$i_1 + i_2 + i_3 = 0$$

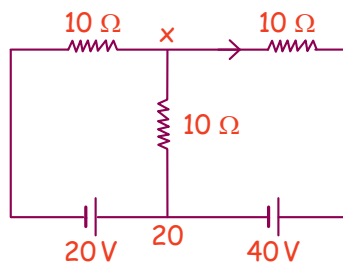
$$\frac{x - 10}{10} + \frac{x - 20}{10} + \frac{x - 30}{10} = 0$$

$$x = 20$$

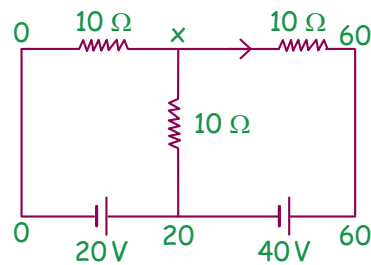
now we can find current though any wire

$$i_1 = \frac{x - 10}{10} = 1A \text{ and } i_2 = \frac{x - 20}{10} = 0$$

Q. Find current through each resistors.



Sol.

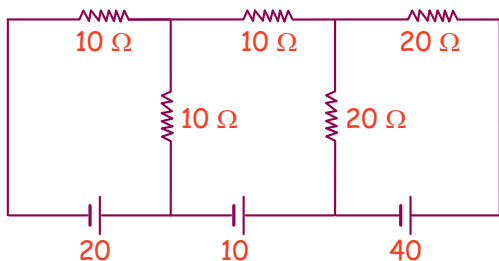


$$i_1 + i_2 + i_3 = 0$$

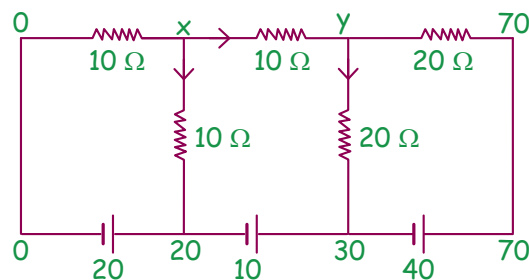
$$\frac{x - 0}{10} + \frac{x - 20}{10} + \frac{x - 60}{10} = 0$$

$$x = \frac{80}{3}$$

Q. Find potential (x).



Sol.



$$\frac{x - 0}{10} + \frac{x - 20}{10} + \frac{x - y}{10} = 0$$

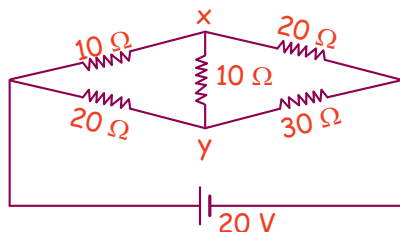
$$3x - y = 20 \quad \dots(1)$$

$$\frac{y - x}{10} + \frac{y - 30}{20} + \frac{y - 70}{20} = 0$$

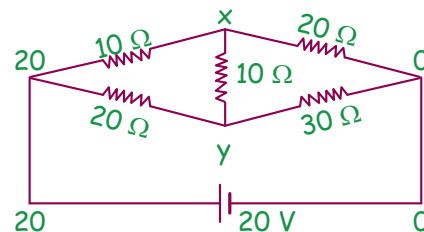
$$4y - 2x = 100 \quad \dots(2)$$

Now we solve both eqn. and get answer

Q. Find $V_x - V_y = ?$



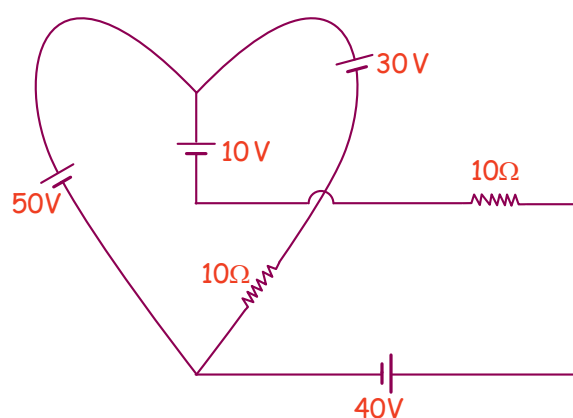
Sol.



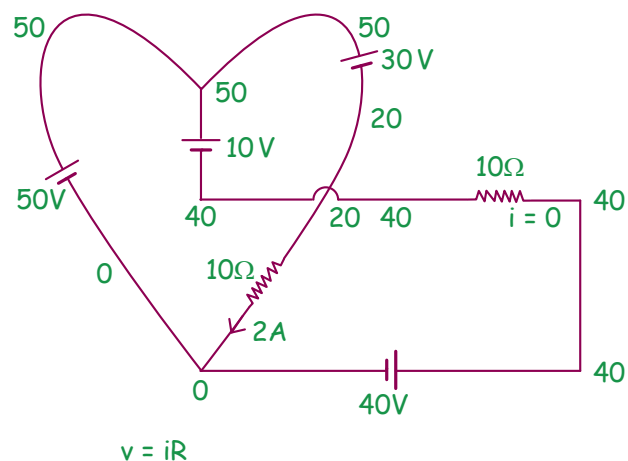
$$\frac{x - 20}{10} + \frac{x - y}{10} + \frac{x - 0}{20} = 0$$

$$\frac{y - 20}{20} + \frac{y - x}{10} + \frac{y - 0}{30} = 0$$

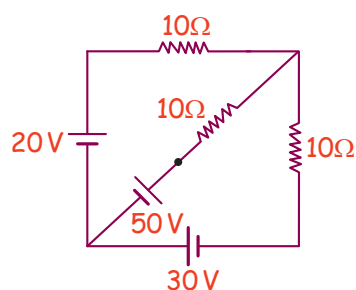
Q. Find current through each resistors.



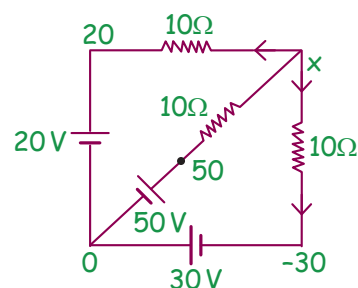
Sol. →



Q. Find current through each resistors.



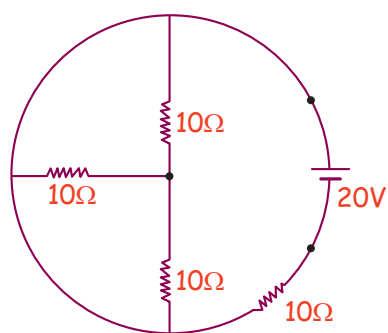
Sol. →



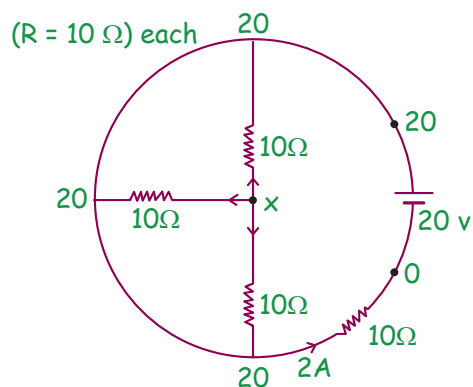
$$\frac{x - 20}{10} + \frac{x - 50}{10} + \frac{x + 30}{10} = 0$$

$$x = \frac{40}{3}$$

Q. Find current through each resistors.



Sol. →

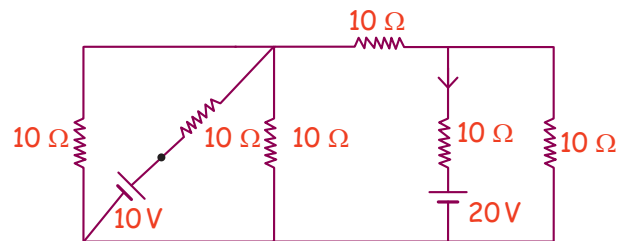


$$\frac{x - 20}{10} + \frac{x - 20}{10} + \frac{x - 20}{10} = 0$$

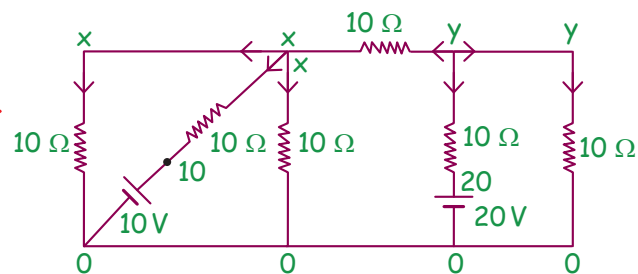
$$x = 20$$

$$i_1 = i_2 = i_3 = 0$$

Q. Find current through each resistors.



Sol. →



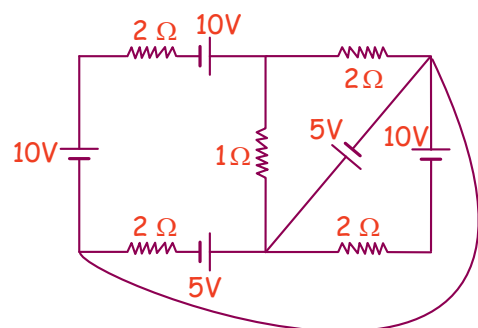
Apply junction law at x

$$\frac{x-0}{10} + \frac{x-10}{10} + \frac{x-0}{10} + \frac{x-y}{10} = 0$$

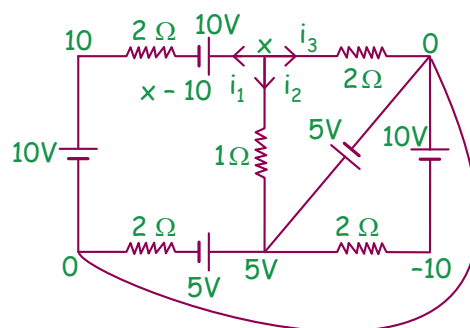
Apply junction law at y

$$\frac{y-x}{10} + \frac{y-20}{10} + \frac{y-0}{10} = 0$$

Q. Find current through each resistors.



Sol. →



$$\frac{x-10-10}{2} + \frac{x-5}{1} + \frac{x-0}{2} = 0$$

$$x - 20 + 2x - 10 + x = 0$$

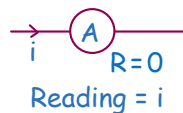
$$4x = 30 \Rightarrow 7.5$$

$$i_2 = \frac{x-5}{1} = 7.5 - 5 = 2.5$$

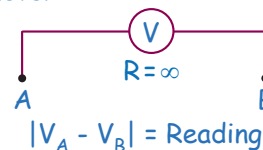


#SKC
ideal Amm ka Resistance
⇒ Zero
ideal Voltmeter ka Resistance ⇒ ∞
Kyonki hum chahte hain ki (A)
(V) ka istemaal karte waqt
circuit na badle.

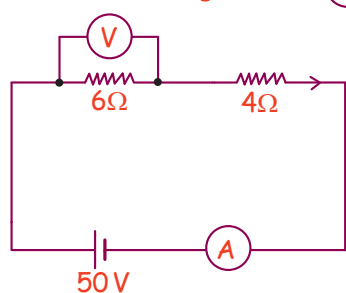
✦ Ideal Ammeter



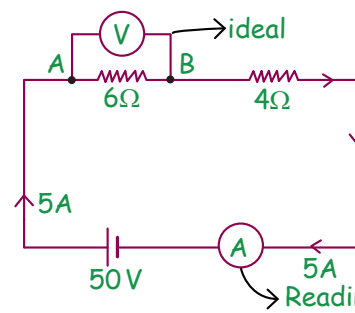
✦ Ideal Voltmeter



Q. Find the reading of ideal (V) & (A)

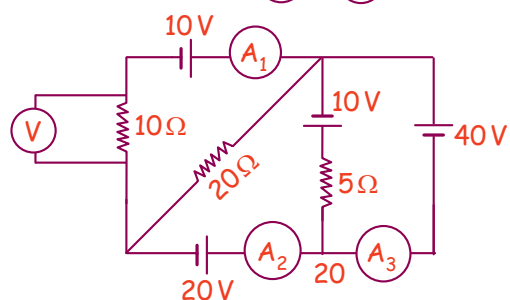


Sol. →



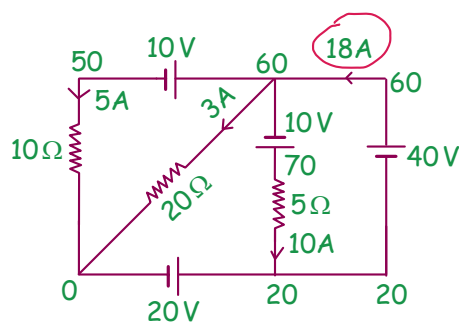
$$V_A - V_B = 5 \times 6 = \text{Reading of voltmeter}$$

Q. Find the reading of A & V



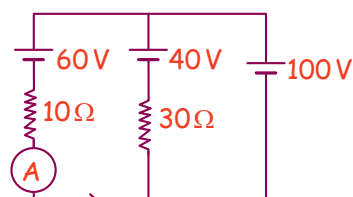
Ammeter ki reading matlab usse kitna current pass kar raha hai agar ammeter ideal hai to use wire maan lo. Voltmeter ki reading matlab jin do point ke beech use connect kiya hai unke beech potential difference. Agar voltmeter ideal hai to uske through $i = 0$.

Sol.

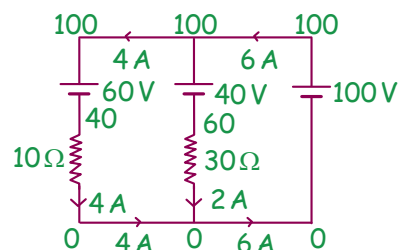


Reading of $A_1 = 5A$
 Reading of $A_2 = 8A$
 Reading of $A_3 = 18A$
 Reading of Voltmeter = 50V

Q. Find the reading of A

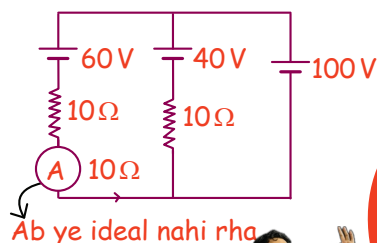


Sol.



Ans. 4 A

Q. What is the reading of non-ideal ammeter?

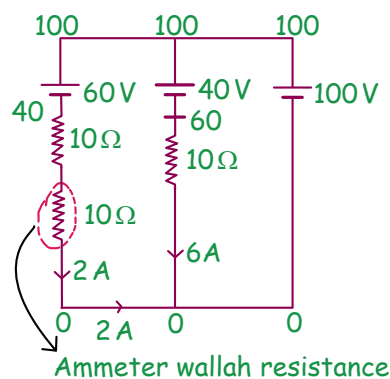


Ab ye ideal nahi rha



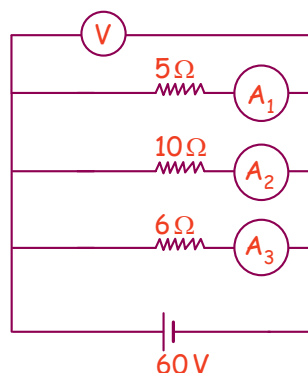
Sol.

#SKC
 agar A - V
 ka Resistance given ho toh
 ye Non-ideal hai
 toh A - V ko
 hatao aur Resistance
 rakhdo

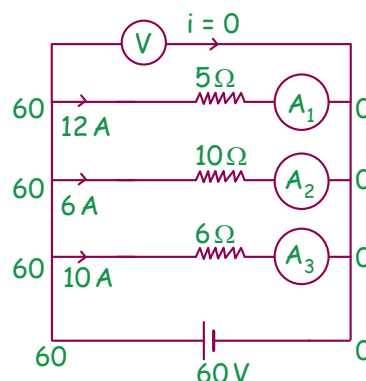


Ammeter wallah resistance

Q. What will be the readings?



Sol.

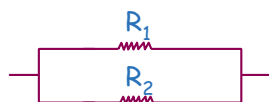


All are ideal:
 Reading of $V = 60V$
 Reading of $A_1 = 12A$
 Reading of $A_2 = 6A$
 Reading of $A_3 = 10A$

If two resistance R_1 and R_2 are in parallel then

Parallel

$\Delta V \rightarrow$ same

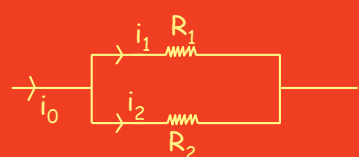


$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{R_2 + R_1}{R_1 R_2}$$

$$R_{eq} = \frac{R_1 R_2}{R_2 + R_1} = \frac{\text{multiply}}{\text{sum}}$$

Q. Sol. $R_{eq} = \frac{10 \times 6}{10 + 6} = \frac{60}{16}$

#SKC



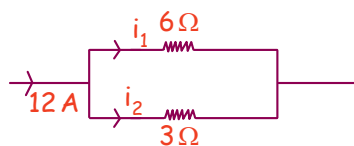
IMP

$i_1 = \text{samne waala 'R' } \times \text{ total current / total R}$

$$i_1 = \frac{R_2}{R_1 + R_2} \times i_0$$

$$i_2 = \frac{R_1}{R_1 + R_2} \times i_0$$

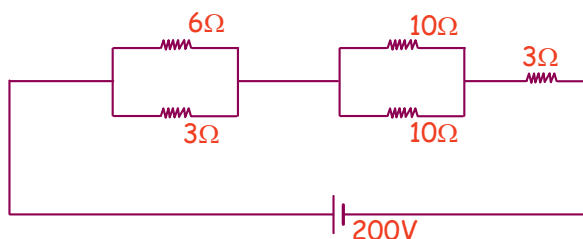
Q. Find i_1 and i_2



$$i_1 = \frac{3}{9} \times 12 = 4A$$

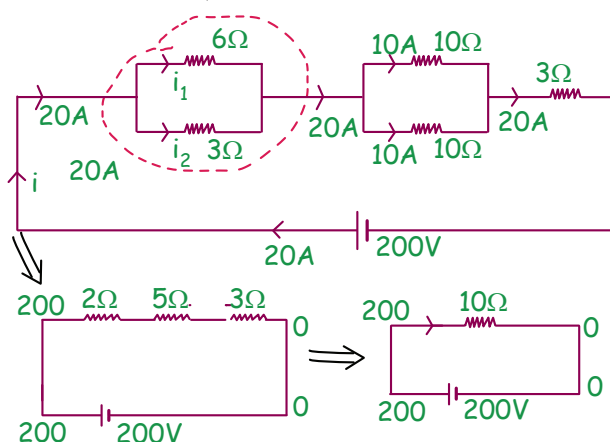
$$i_2 = \frac{6}{9} \times 12 = 8A$$

Q. Find current through each resistors.



Sol.

$$R_{eq} = \frac{\text{Multiply}}{\text{sum}}$$



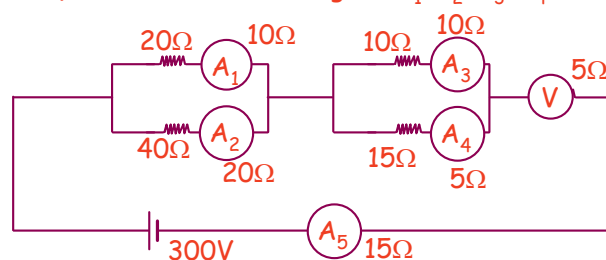
$$R_{eq} = 2 + 5 + 3 = 10\Omega$$

$$i = \frac{V}{R_{eq}} = \frac{200}{10} = 20A$$

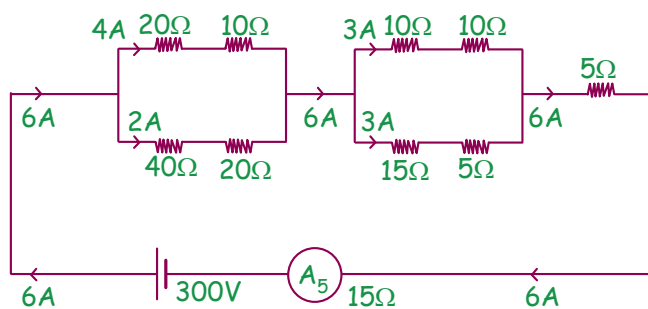
$$i_1 = \frac{3}{3+6} \times 20 = \frac{20}{3}$$

$$i_2 = \frac{6}{3+6} \times 20 = \frac{120}{9}$$

SSSQ. What will be reading of A_1 A_2 A_3 A_4 and V ?



Sol.



Reading of $A_1 \rightarrow 4A$

Reading of $A_2 \rightarrow 2A$

Reading of $A_3 \rightarrow 3A$

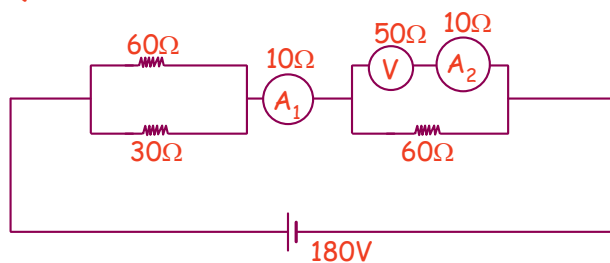
Reading of $A_4 \rightarrow 3A$

Reading of $A_5 \rightarrow 6A$

Reading of voltmeter = $6 \times 5 = 30$

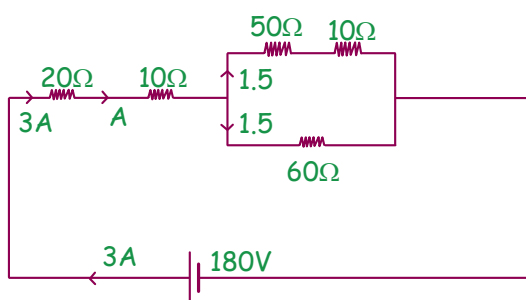
SSSQ. Saleem Sir Special Question

Q.



If reading A_1 , A_2 & V are x , y , z , find $\frac{z}{xy}$

Sol.



$$R_{eq} = 20 + 10 + 30 = 60\Omega$$

$$A_1 \rightarrow 3A = x$$

$$A_2 \rightarrow 1.5A = y$$

$$V \rightarrow 15 \text{ volt} = z$$

$$\frac{z}{xy} = \frac{15}{3 \times 1.5} = 3.33$$



Ab hum R_{eq} nikalna sikhege

★ If resistance are in series then

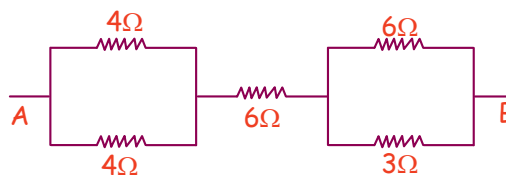
$$R_{eq} = R_1 + R_2 + R_3 + \dots$$

★ If resistance in parallel

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

★ If R_1 and R_2 are in parallel then $R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$

Q. Find R_{eq}

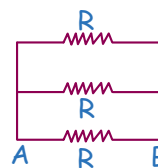
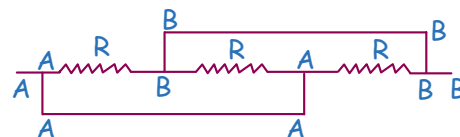


$$R_{eq} = 2 + 6 + 2 = 10\Omega$$

Q. Find R_{eq} between A and B.

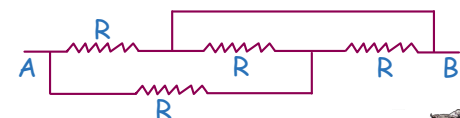


Sol.

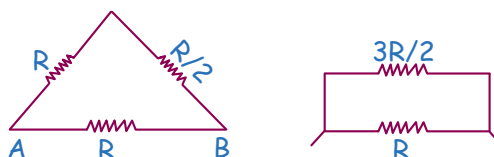
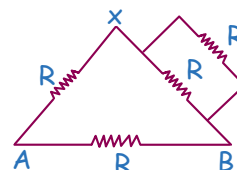
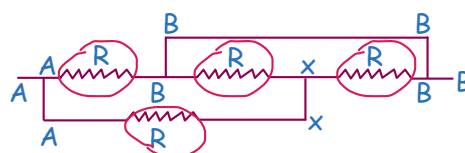


$$\frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} = \frac{3}{R} \quad R_{eq} = R/3$$

Q. Find R_{eq} between A and B.

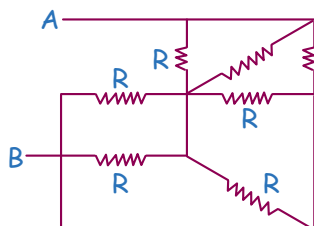


Sol. Jahaan phaso wahin x ko yaad karo aur uske kharaab, bekaar, daraawani soorat ko sudhaar lo... Abe matlab naya circuit diagram banao.

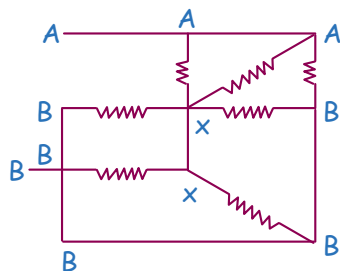


$$\text{Ans. } R_{eq} = 0.6R$$

Q. Find the R_{eq} between A and B.

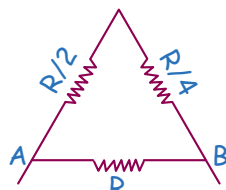
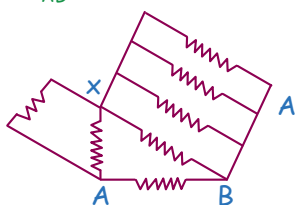


Sol. Resistance $R\Omega$ each



All resistance are equal ($R\Omega$)

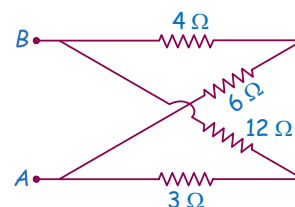
$R_{AB} = ?$



$$R_{eq} = \frac{\frac{3R}{4} \times R}{\frac{3R}{4} + R} = \frac{3}{7}R$$

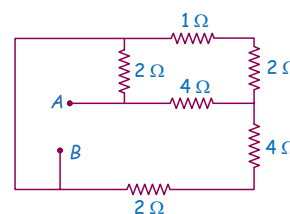
Homework

Q. In the given network, the equivalent resistance between A and B is



Ans. 5

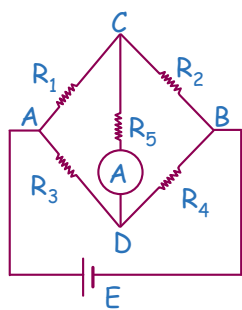
Q. In the circuit shown in figure, equivalent resistance between A and B is



Ans. 1.5

Wheatstone Bridge

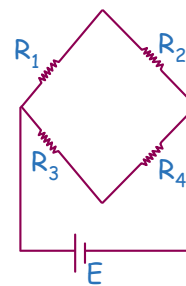
☆ In following circuit



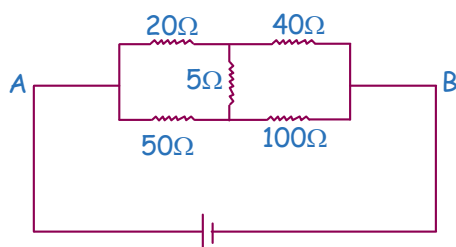
If $\frac{R_1}{R_2} = \frac{R_3}{R_4}$ then be observed

that $V_C = V_D$ and current through R_5 is zero to R_5 ko circuit se udha do such type of circuit called balance wheatstone bridge

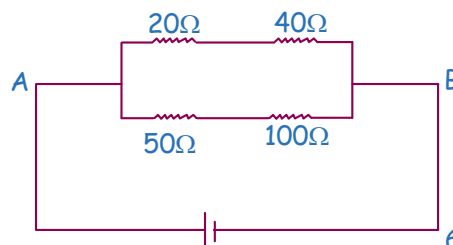
$\Rightarrow$



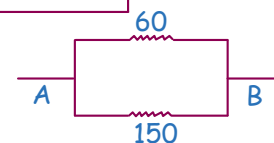
Q. Find R_{eq} between A & B



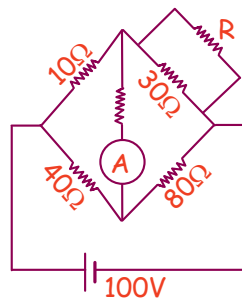
Sol.



$$R_{AB} = \frac{60 \times 150}{210}$$

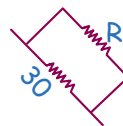


Q. Find value of R for which ammeter reading is zero



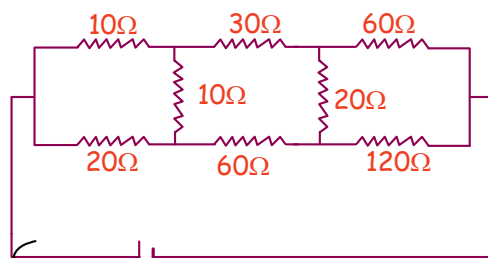
Sol.

$$R = 60$$

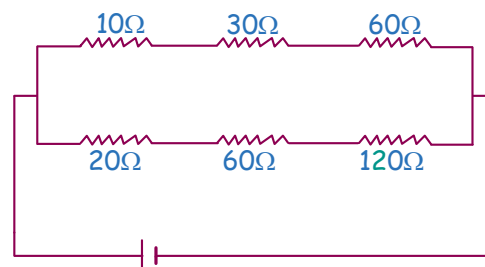


$\Rightarrow 20\Omega$ hona chahiye

Q. Find equivalent resistance.

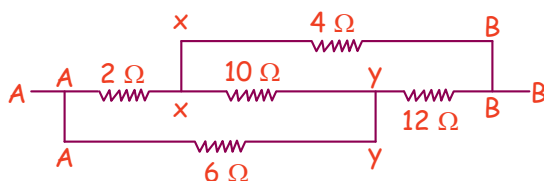


Sol.

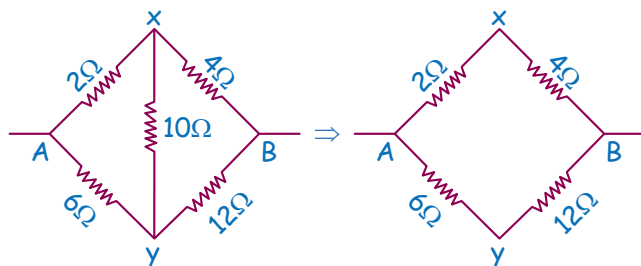


isme wheatstone Bridge chupa hai

Q. find $R_{AB} = ?$



Sol.



$$R_{eq} = \frac{6 \times 18}{6 + 18} = \frac{6 \times 18}{24} = 4.5$$

Kirchhoff Voltage Law (KVL)

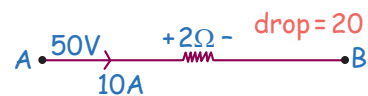


$$V_A - V_B = 10$$



$$V_A - V_B = iR$$

$$V_A - iR = V_B$$



$$V_A - iR = V_B$$

$$50 - 10 \times 2 = V_B$$

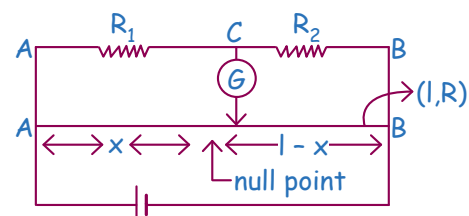
$$V_B = 30$$

Meter Bridge

- It is used to find unknown resistance.
- Its concept based on balance wheat stone bridge.
- When current through galvanometer is Zero.

$$\frac{R_1}{R_2} = \frac{R_{AD}}{R_{DB}} = \frac{x}{l-x} \quad (l = 100 \text{ cm})$$

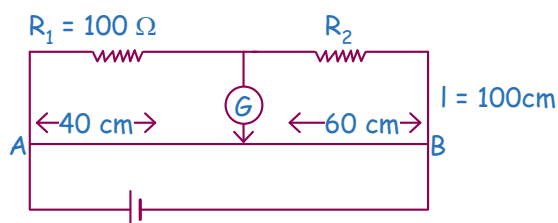
$$\frac{R_1}{R_2} = \frac{x}{l-x}$$



$$R = \frac{\rho l}{A}$$

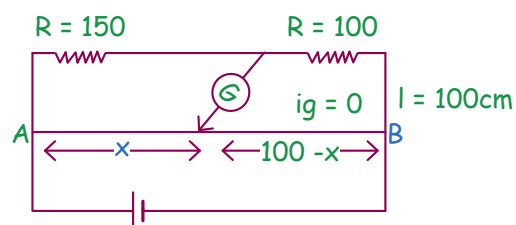
$$R \propto \text{length}$$

Q. In a meter bridge (as shown in fig), if null point is found at a distance of 40 cm from A. Find shift in the null point if R_1 & R_2 interchanged



Sol. $\frac{R_1}{R_2} = \frac{40}{60}$

$\frac{100}{R_2} = \frac{40}{60} \Rightarrow R_2 = 150$



$\frac{150}{100} = \frac{x}{100 - x}$

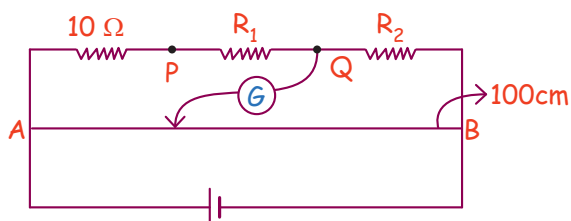
$\frac{3}{2} = \frac{x}{100 - x}$

$300 - 3x = 2x$

$x = 60$

Ans. shift = $60 - 40 = 20$ cm

Q. When galvanometer is connected to point P, null point is at a distance 40 cm. From point A when galvanometer is connected to point Q, null point is found at a distance 30 cm from end B. Find R_1 & R_2 .

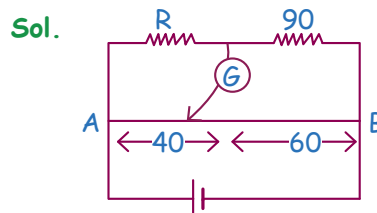


Sol. $\frac{10}{R_1 + R_2} = \frac{40}{60}$... (1)

$\frac{10 + R_1}{R_2} = \frac{70}{30}$... (2)

Solve and get $R_1 = R_2 = 7.5 \Omega$

Q. In a meter Bridge find value of R if end correction are 1 cm & 3 cm at end A & end B



$\frac{R}{90} = \frac{40 + 1}{60 + 3}$

POWER DISSIPATED ACROSS RESISTANCE



✦ Power dissipated across the resistance = $i^2 R$

✦ $P = V \cdot i = iR \cdot i = i^2 R = \frac{V^2}{R}$

✦ $H = \int_0^t i^2 R dt$

✦ If current is constant heat = $i^2 R t$

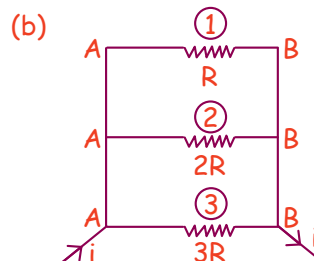
Q. Find the order of power dissipated across the resistance in following question.



Sol. $P = i^2 R$

$R \uparrow = P \uparrow$ Power loss $\uparrow$

$P_3 > P_2 > P_1$

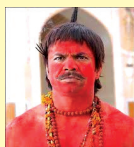


Sol. $V \rightarrow$ same

$P = \frac{V^2}{R}$

$R \uparrow = P \downarrow$

$P_3 < P_2 < P_1$



Student: sir ye batao
ki kab $P = i^2 R$ lagaana
hai kab $P = v^2/R$
lagaana hai



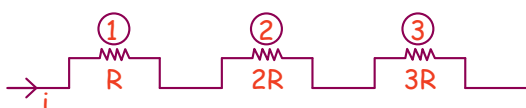
Saleem
sir be
like

Dono me se jo chahe
formula laga do ans. same
aayega.

**kya re tu, naak mein dum
kar rakha hai tune**

Phir bhi agar resistance series mein hai to $P = i^2 R$
try karo aur agar resistance parallel mein hai toh
 $P = V^2/R$ try karo calculation aasan rhegi

Q. Three different bulb of resistance R , $2R$, $3R$
are in series as shown in figure. Find order of
their brightness.



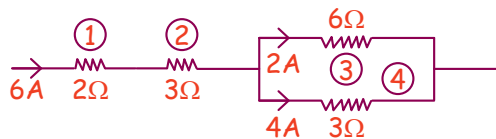
Sol. $i \rightarrow$ same $P = i^2 R$

$$P \rightarrow P_3 > P_2 > P_1$$

$$\text{Brig} \rightarrow B_3 > B_2 > B_1$$

Bulb ke case mein jiske across power
 $\uparrow$ toh Brightness $\uparrow$

Q. Repeat the above problem in following case.



Sol. $P_1 = 6^2 \times 2 = 72$

$$P_2 = 6^2 \times 3 = 108$$

$$P_3 = 2^2 \times 6 = 24$$

$$P_4 = 4^2 \times 3 = 48$$

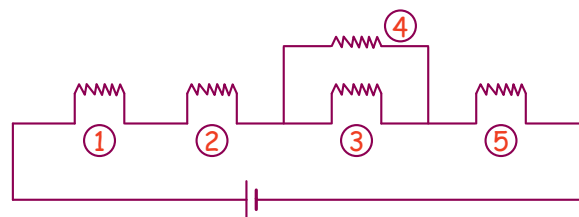
$$P_2 > P_1 > P_4 > P_3$$

$$\text{Brightness } B_2 > B_1 > B_4 > B_3$$

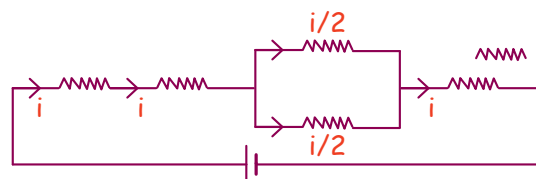
Q. Compare brighten of bulb

All are identical bulb

$R \rightarrow$ same

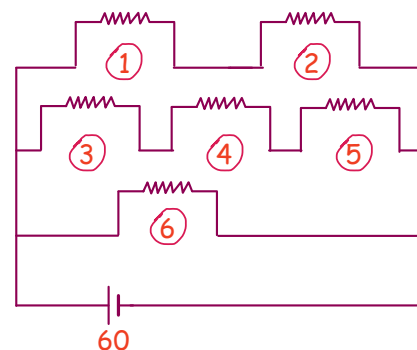


Sol.

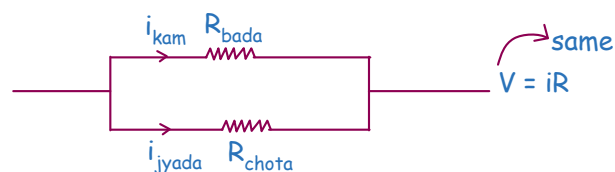


$$B_1 = B_2 = B_5 > B_4 = B_3$$

Q. All bulbs are identical having same resistance
 10Ω . Find order of brightness

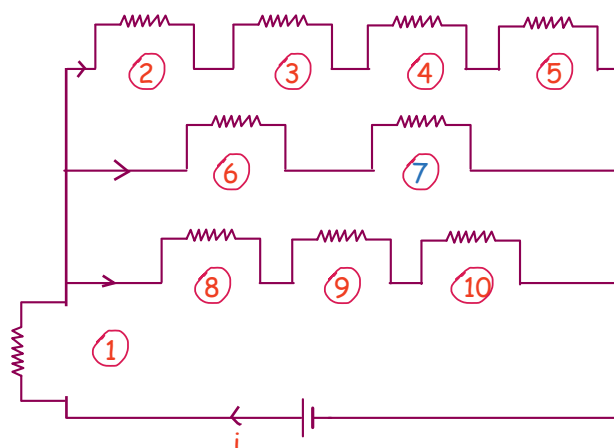


Sol. $B_6 > B_1 = B_2 > B_3 = B_4 = B_5$



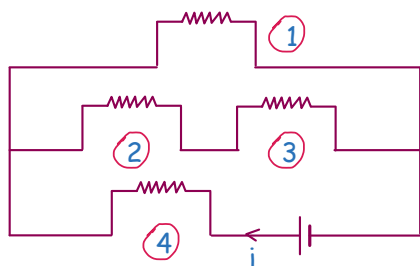
Q. All are identical

(a)



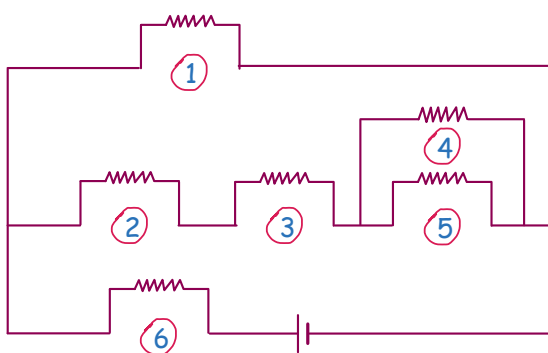
Sol. $B_1 > B_6 = B_7 > B_8 = B_9 = B_{10} > B_2 = B_3 = B_4 = B_5$

(b) All bulbs are identical



Sol. $B_4 > B_1 > B_2 = B_3$

(c)



Sol. $B_6 > B_1 > B_2 = B_3 > B_4 = B_5$

Bulb

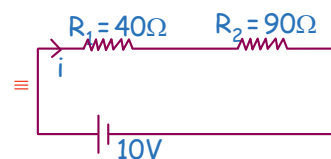
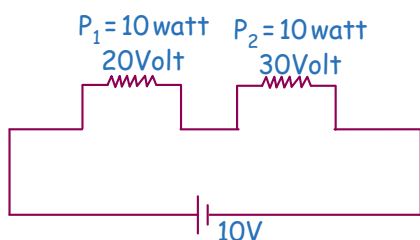


Suppose it's given that rated voltage of bulb is V and rated power is P it means yeh bulb ka resistance dene

ka trika hai

$$\text{Resistance of bulb} = \frac{V^2}{P}$$

Q. Compare brighter of bulbs.



$$\text{Sol. } R = \frac{V^2}{P} \Rightarrow R_1 = \frac{(20)^2}{10} = 40\Omega$$

$$i = \frac{10}{130}$$

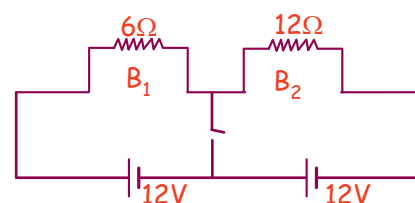
$$R_2 = \frac{(30)^2}{10} = 90\Omega$$

$$\text{Power across } B_1 = i^2 R_1 = (1/13)^2 \times 40$$

$$B_2 = i^2 R_2 = (1/13)^2 \times 90$$

$B_2 > B_1$ (or $i \rightarrow \text{same} \Rightarrow \text{jiska } R \uparrow \text{ uski } P \uparrow B \uparrow$).

Q. B_1 and B_2 are bulbs. What happens to brightness of bulbs after switch close?



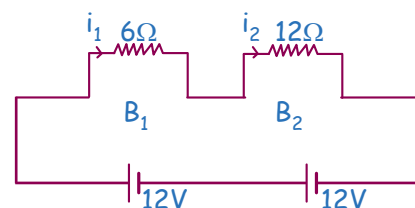
1. Brightness of B_1 inc

2. Brightness of B_1 dec

3. Brightness of B_2 inc

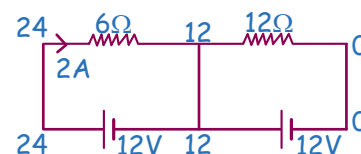
4. Brightness of B_2 dec

Sol. Before switch is close



$$i_1 = i_2 = 1.33$$

Now after switch is closed.



$$i_1 = 2 \text{ and } i_2 = 1$$

We observed that after switch is closed current through B_1 increased hence power across bulb B_1 increased similarly current across bulb 2 decrease. Hence power across bulb 2 decrease.

Ans: 1 & 4

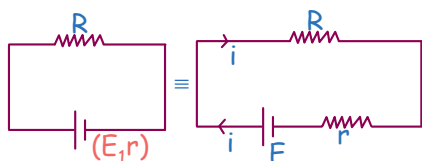
- When connected voltage and rated voltage are same then total power dissipated for n number of bulbs in series,

$$\frac{1}{P_T} = \frac{1}{P_1} + \frac{1}{P_2} + \frac{1}{P_3} + \dots + \frac{1}{P_n}$$

- When connected voltage and rated voltage are same then total power dissipated for n number of bulbs in parallel,

$$P_T = P_1 + P_2 + P_3 + \dots + P_n$$

Max Power Theorem



$$\text{Power dissipated across } R = i^2 R = \left(\frac{E}{R+r} \right)^2 R$$

$$P = \frac{E^2 R}{(R+r)^2}$$

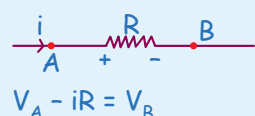
At what condition $P \rightarrow \max$

$$\text{Do } \frac{dP}{dR} = 0 \Rightarrow \text{Solve and get } R = r$$

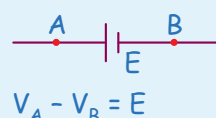
So maximum power will be delivered to load R if value is equal to that of internal resistance of cell. In above case $P_{\max} = E^2/4R$ (after solve).



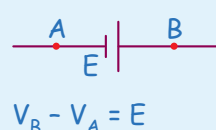
Ab hum KVL likhna sikhenge jiska apko bahut der se intazaar hai bas neeche ki baate yaad rakho.



(Resistance mein current ki disha mein iR ka drop)

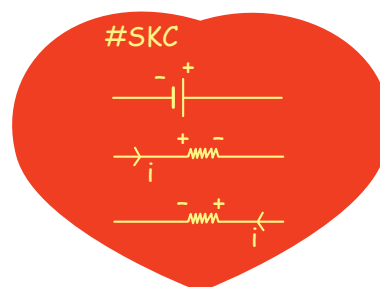


(irrespective of dirⁿ of current)



(chahe current ho ya na ho khi bhi ho)

Q. Find $V_A - V_B$



Sol.



$$V_A - iR + E = V_B$$

$$V_A - 20 + 30 = V_B$$

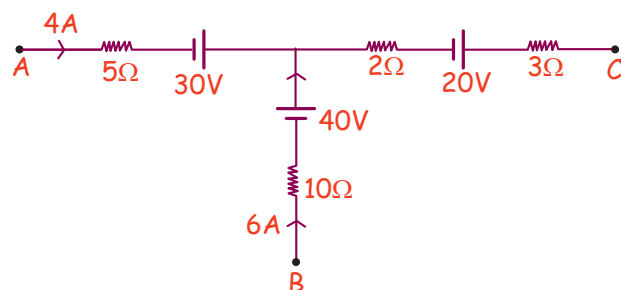
$$V_A - V_B = -10$$



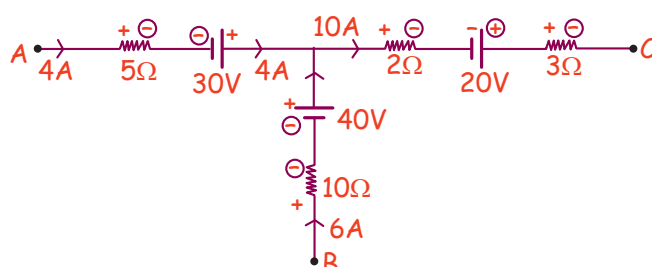
Sol. $V_A - E_1 - iR_1 + E_2 - iR_2 - E_3 - iR_3 = V_B$

$$V_A - V_B = \checkmark$$

Q. Find (1) $V_A - V_C$ (2) $V_A - V_B$ (3) $V_B - V_C$



Sol.



(1) $V_A - V_C = ?$

$$V_A - 20 + 30 - 10 \times 2 + 20 - 30 = +V_C$$

$$V_A - V_C = 20$$

$$(2) V_A - V_B = ?$$

$$V_A - 20 + 30 - 40 + 6 \times 10 = V_B$$

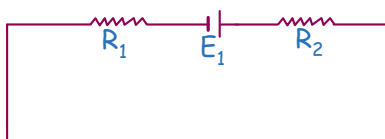
$$V_A - V_B = -30$$

$$(3) V_B - V_C = ?$$

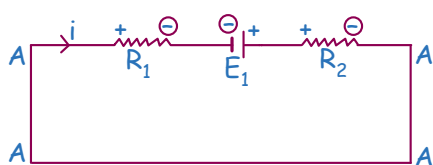
$$V_B - 60 + 40 - 20 + 20 - 30 = V_C$$

$$V_B - V_C = 50$$

Q. Find current in each resistors.



Sol.

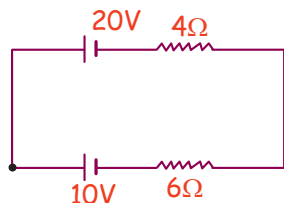


$$V_A - iR_1 + E_1 - iR_2 = V_A$$

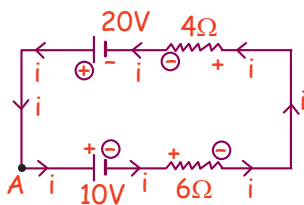
$$E_1 = iR_1 + iR_2$$

$$i = \frac{E_1}{R_1 + R_2}$$

Q. Find current in the circuit.



Sol.



$$V_A - 10 - 6i - i \times 4 + 20 = V_A$$

$$-10 - 10i + 20 = 0$$

$$i = 1A$$

thodi der ke badh
hum dekhege
20 V battery is
discharging and
10 V batter is
charging in this
case.

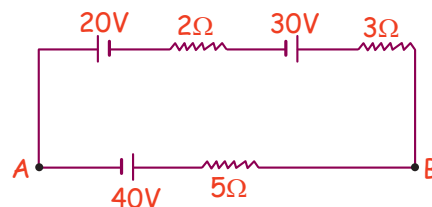
#SKC

$i \rightarrow$ loop mein $C\omega/A\omega$
Jo dil chahe maan lo agar
galat.....maani toh
.....aa jayega

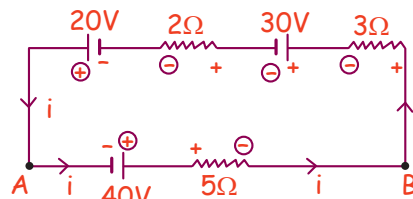


Q. (1) Find current in circuit

$$(2) V_A - V_B = ?$$



Sol.



(1) $i \rightarrow$ Acw maan lo (let)

$$V_A + 40 - 5i - 3i - 30 - 2i + 20 = V_A$$

$$i = 3A$$

(2) $V_A + 40 - 3 \times 5 = V_B$

$$V_A - V_B = -25$$



1. A se B chalte hai

$$V_A - 30 + 2 \times 10 = V_B$$

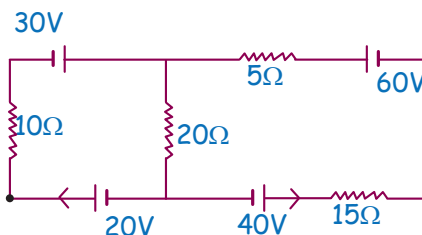
$$V_A - V_B = 10V$$

2. B se A chalte hai

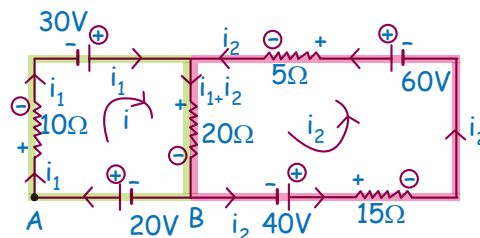
$$V_B - 2 \times 10 + 30 = V_A$$

$$V_A - V_B = 10$$

Q. Find current in each in each resistors



Sol.



$$V_A - 10i + 30 - (i_1 + i_2) \times 20 + 20 = V_A$$

$$-10i_1 + 30 - (i_1 + i_2) \times 20 + 20 = 0 \quad \dots(1)$$

$$V_B + 40 - 15i_2 + 60 - 5i_2 - (i_1 + i_2) \times 20 = V_B$$

$$40 - 15i_2 + 60 - 5i_2 - (i_1 + i_2) \times 20 = 0 \quad \dots(2)$$

Solve (1) & (2) And get Ans:

Charging & Discharging of Batteries

#SKC

Agar battery ke bade dande se baahar current nikal rahaa hai to battery discharge ho rahee hai aur bade dande ke andar current aa rahaa hai to battery charge ho rahee hai



Discharge of battery



Charging of battery

Charging of battery



$$V_A - E - ir = V_B$$

$$V_A - V_B = E + ir$$

(Pot diff across battery, terminal voltage)

Discharging of battery



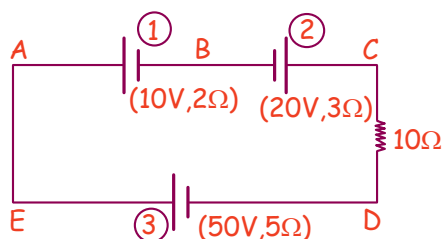
$$V_A - E + ir = V_B$$

$$V_A - V_B = E - ir$$

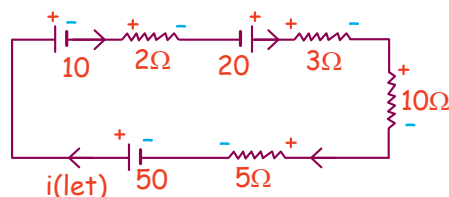
(Pot diff across battery, terminal voltage)

Ab pichhle page mein SKC ke paas waale questions solve karo aur dekho kaunsi battery charge ho rahee hai aur kaunsi discharge.

Q. Find potential difference across each battery



Sol.



$$V_A - 10 - 2i + 20V - 3i - 10i - 5i + 50 = V_A$$

$$60 = 2i$$

$$i = 3A$$

(1) → Charging

2, 3 → discharge

Potential diff across each battery

$$|V_{AB}| = E + ir = 10 + 3 \times 2 = 16V$$

$$|V_{BC}| = |E - ir| = |20 - 3 \times 3| = 11V$$

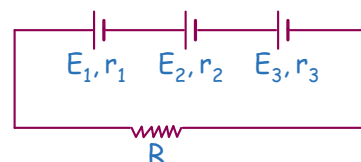
$$|V_{ED}| = |50 - 3 \times 5| = 35$$

#SKC

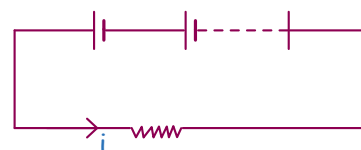
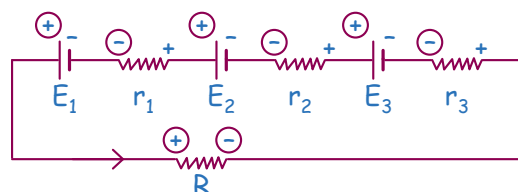
Battery ke internal resistance se bilkul nahi darna hai bas battery ke baju mein utana resistance lga dena hai



Q. Find current in circuit



Sol.

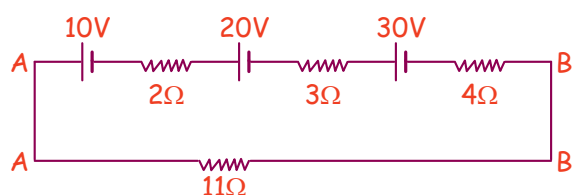


$$-iR - iR_3 + E_3 - iR_2 + E_2 - iR_1 + E_1 = 0$$

$$i = \frac{E_1 + E_2 + E_3}{r_1 + r_2 + r_3 + R}$$

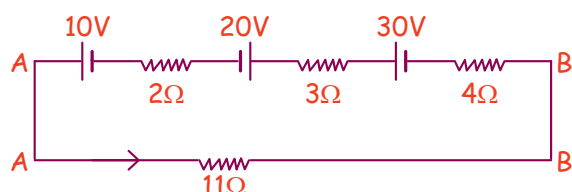
$$i = \frac{E_1 + E_2 + \dots}{(r_1 + r_2 + \dots) + R}$$

Q. Find current in circuit



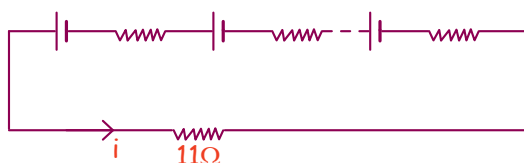
Sol. $i = \frac{10 + 20 + 30}{2 + 3 + 4 + 11}$ $V_{AB} = \checkmark$

Q. Find current in circuit



Sol. $i = \frac{10 - 20 + 30}{2 + 3 + 4 + 11}$

Q. (a) n identical cells are connected in series as shown in diagram. Find i through R



Sol. $i = \frac{E + E + \dots}{(r + r + r \dots)R} = \frac{nE}{nr + R}$

(b) If one cell is reversed find current in R

Sol. $i = \frac{nE - 2E}{nr + R}$

(c) If m battery reversed

Sol. $i = \frac{nE - 2mE}{nr + R} = \frac{(n - 2m)E}{nr + R}$

Results

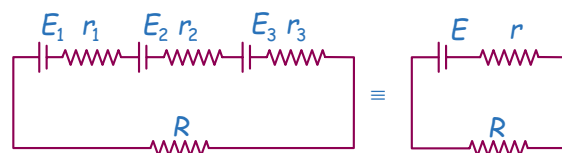
(i) Series Combination

If n sources of emf are connected in series with same polarity, then the equivalent emf is given by

$$E = E_1 + E_2 + E_3 + \dots + E_n$$

And, total internal resistance is

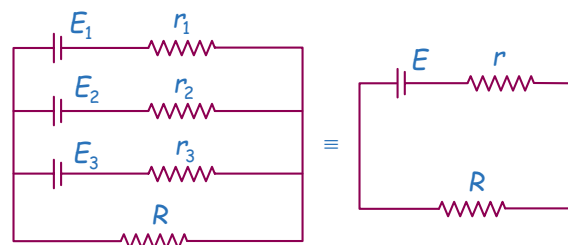
$$r = r_1 + r_2 + r_3 + \dots + r_n$$



✦ If there are ' n ' identical cells with emf E and internal resistance ' r ' and they are connected in such a way that p cells are connected in opposite polarity then,

$$E_{\text{net}} = (n - 2p)E \text{ and } r_{\text{net}} = nr$$

(ii) Parallel Combination



The emf and internal resistance of the equivalent battery are given by

$$E = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2} + \frac{E_3}{r_3}}{\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}}$$

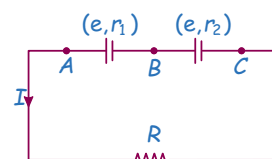
$$\text{and } \frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$$

mere sapne mein aai sundar pari
And niche vala questions
is compulsory



Q. Two batteries having the same emf ε but different internal resistances r_1 and r_2 are connected in series in same polarity with an external resistor R . For what value of R does the potential difference between the terminals of the first battery become zero?

Sol.



Net resistance in the circuit is $(r_1 + r_2 + R)$.

Current in the circuit

$$I = \frac{2\varepsilon}{(r_1 + r_2 + R)}$$

The potential difference between the terminals of first battery is $(V_A - V_B)$. Terminal potential difference is given by

$$(V_A - V_B) = \varepsilon - Ir_1$$

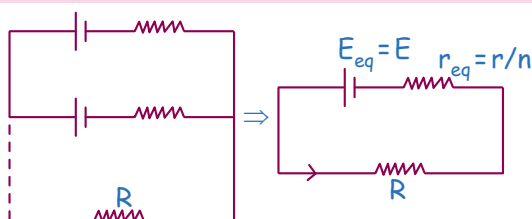
$$V_A - V_B = \varepsilon - \frac{2\varepsilon r_1}{r_1 + r_2 + R} = \varepsilon \frac{(R + r_2 - r_1)}{(R + r_2 + r_1)}$$

For $(V_A - V_B)$ to be zero, we must have

$$R = (r_1 - r_2)$$

This gives meaningful result only if $r_1 > r_2$. Otherwise, if $r_2 > r_1$, then $R = r_2 - r_1$ will produce terminal voltage across second cell to be zero ($V_{BC} = 0$).

★ If n identical (E, r) cells are in parallel



$$i = \frac{E}{\frac{r}{n} + R}$$

$$E_{eq} = \frac{E/r + E/r + \dots n \text{ times}}{1/r + 1/r + \dots n \text{ times}}$$

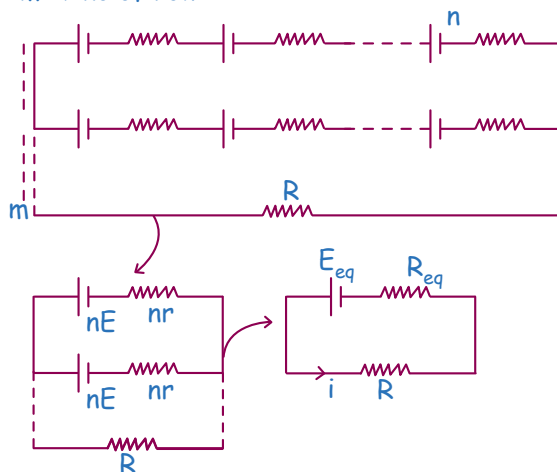
$$E_{eq} = E$$

$$\frac{1}{R_{eq}} = \frac{1}{r} + \frac{1}{r} + \dots n \text{ times}$$

$$r_{eq} = \frac{r}{n}$$

★ Mixed Grouping

n identical cell in row in series
 $m \rightarrow$ no of row



$$\text{net emf} = nE$$

$$\text{Total internal resistance} = \frac{nr}{m}$$

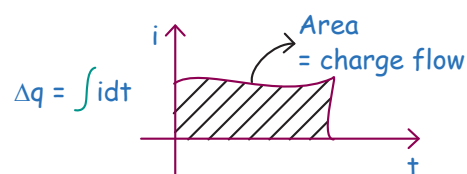
$$i = \frac{nE}{\frac{nr}{m} + R}$$

Ab ham padhenge vo cheez jo saari duniya is chapter mein sabse pehle padhti hai.



Current Electricity

$$\text{Current} = i = \frac{dq}{dt} = \text{Rate of flow of charge}$$



Conventionally direction of flow of current is taken to be in direction of flow of +ve charge

If moving charge is negative (jaise electron) current direction is opposite to direction of motion of charge.



Current Density

★ It is the current flowing per unit area normal to the surface

★ vector quantity

★ Flux of current density is current

★ It's direction is same as direction of current



$$i = \vec{J} \cdot \vec{A} = JA \cos \theta$$

$$\Rightarrow J = \frac{i}{A \cos \theta}$$

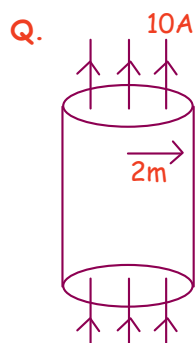
For special case

$$\theta = 0$$



$$i = JA \Rightarrow J = \frac{i}{A}$$

$$i = \int \vec{J} \cdot d\vec{A} \quad \text{agar } J \text{ badla toh we will use this}$$

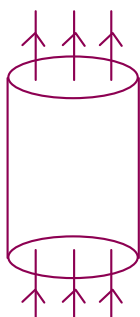


Sol. $\vec{J} = \frac{10}{\pi 2^2} \hat{i}$

$\phi = \int \vec{E} \cdot d\vec{A}$

Current density
का flux = $\int \vec{J} \cdot d\vec{A} = i$

Q. If $J = J_0 r$, Find total current flowing through long cylindrical wire

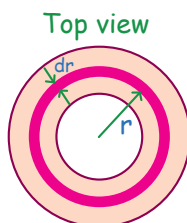


Sol. r par jaake dr thickness ka ek hollow cylinder pakda suppose isse di current pass kiya

$di = \vec{J} \cdot d\vec{A}$

$\int di = \int_0^R J_0 r 2\pi r \cdot dr$

$i = J_0 2\pi \frac{R^3}{3}$



MOTION OF ELECTRON INSIDE CONDUCTOR

In absence of applied potential difference electrons have random motion.

All the free electrons are in random motion due to the thermal energy and follow the relationship given by

$\frac{3}{2} k_B T = \frac{1}{2} m v^2$

where, k_B = Boltzmann's constant

At room temperature their speed is around 10^6 m/s but the average velocity is zero, so net current is also zero.

Mean Free Path

The average distance travelled by a free electron between two consecutive collisions is called as mean free path λ .

Mean free path $\lambda = \frac{\text{total distance travelled}}{\text{number of collisions}}$

Relaxation Time

The time taken by an electron between two successive collisions is called as relaxation time τ . ($\tau \sim 10^{-14}$ s)

Relaxation time $\tau = \frac{\text{total time taken}}{\text{number of collisions}}$

The thermal speed can be written as $v_T = \frac{\lambda}{\tau}$

Thermal energy Random direct, zigzag path



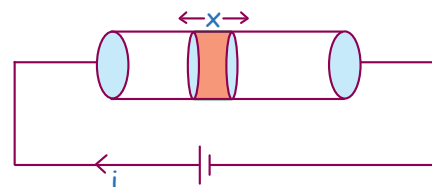
$\vec{u}_1 + \vec{u}_2 + \vec{u}_3 + \dots + \vec{u}_A = 0$

Now battery is applied due which potential different created across conductor result into electric field.

Drift Velocity

- Rate at which random motion of free electron drift in the presence of applied electric field is called velocity (mm/sec order).
- It is the average of velocity of charge carries over the no of charge carrier.
- Drift velocity is v with which the free electron get drifted toward the positive terminal under the effect of applied external Electric field.

Let $n \rightarrow$ no of electron per unit vol^m



$i = \frac{\Delta q}{\Delta t}$ $\Delta q = n A x e$

$i = \frac{n A x e}{x/v_d} = n A e v_d$

Charge on electron

$i = n e A V_d$

Drift velocity

Area of cross section

No. of free e^- per unit vol^m

Derivation for Drift Velocity, J

$$\vec{v}_1 = \vec{u}_1 + \vec{a}\tau_1$$

where, $\vec{a}$ = acceleration of electron = $\frac{-e\vec{E}}{m_e}$

τ = Relaxation time



Similarly for other electrons:

$$\vec{v}_2 = \vec{u}_2 + \vec{a}\tau_2$$

$$\vec{v}_n = \vec{u}_n + \vec{a}\tau_n$$

Average velocity of all the free electrons in the conductor is equal to the drift velocity $\vec{v}_d$ of the free electrons.

$$\begin{aligned}\vec{v}_d &= \frac{\vec{v}_1 + \vec{v}_2 + \vec{v}_3 + \dots + \vec{v}_n}{n} \\ &= \frac{(\vec{u}_1 + \vec{a}\tau_1) + (\vec{u}_2 + \vec{a}\tau_2) + \dots + (\vec{u}_n + \vec{a}\tau_n)}{n} \\ &= \left(\frac{\vec{u}_1 + \vec{u}_2 + \vec{u}_3 + \dots + \vec{u}_n}{n} \right) + \vec{a} \left(\frac{\tau_1 + \tau_2 + \tau_3 + \dots + \tau_n}{n} \right)\end{aligned}$$

Since average thermal speed = 0

$$\text{So, } \frac{\vec{u}_1 + \vec{u}_2 + \vec{u}_3 + \dots + \vec{u}_n}{n} = 0 \text{ and}$$

$$\frac{\tau_1 + \tau_2 + \tau_3 + \dots + \tau_n}{n} = \tau = \text{average relaxation time.}$$

$$\text{So, } \vec{v}_d = \vec{a}\tau \Rightarrow \vec{v}_d = \frac{-e\vec{E}}{m_e}(\tau)$$

Where,

$\vec{v}_d$ = Drift velocity of electrons

$\vec{E}$ = Electric field applied

m_e = Mass of electron

τ = Average relaxation time

The direction of drift velocity for electrons in a metal is opposite to that of applied field $\vec{E}$.

$$i = V_d enA = \left(\frac{eE}{m} \right) \tau enA = \frac{eV}{ml} \tau enA$$



$$i = \frac{e^2 \tau n v A}{ml} = \frac{v}{\left(\frac{ml}{e^2 n A} \right)} = \frac{V}{R} = \frac{\Delta V}{R}$$

$$E = \frac{V}{l} = \frac{\text{Pot. diff}}{\text{length}}$$

$$\Delta V = iR$$

Ohm's Law

and

$$i = V_d enA$$

$$\frac{i}{A} = V_d en = \frac{eE}{m} \tau en$$

$$J = \frac{e^2 \tau n}{m} E = \sigma E$$

$$\vec{J} = \sigma \vec{E}$$

$$\text{mobility} = \mu = \frac{\text{Drift velocity}}{E.F}$$

$$\mu = \frac{V_d}{E}$$

$$\Delta V = iR \text{ (ohm's law)}$$

substance which follow ohm's law called ohmic structure.

$$\frac{1}{\rho} = \sigma = \frac{\tau e^2 n}{m} = \text{conductivity}$$

$$R = \frac{\rho l}{A} = \frac{l}{\sigma A}$$



$$\star V = iR$$

$$\star i = V_d enA$$

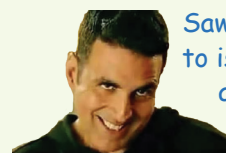
$$\star v_d = \frac{eE}{m} \tau$$

$$\star \vec{J} = \sigma \vec{E}$$

$$\star R = \rho \frac{l}{A} = \frac{l}{\sigma A} \quad \left(\begin{array}{l} \rho = \text{resistivity} \\ \sigma = \text{conductivity} \end{array} \right)$$

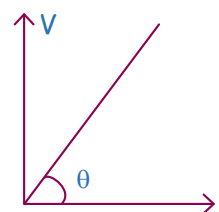
$$\star \frac{1}{\rho} = \sigma = \frac{\tau e^2 n}{m} = \text{conductivity}$$

$$\star \text{mobility} = \mu = \frac{\text{Drift velocity}}{E.F} = \frac{V_d}{E}$$



Sawal aayega to isi BOX se aayega.

★ Ohm's law $i \propto V$



$$V = iR$$

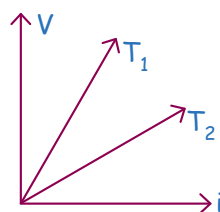
$$y = mn$$

$$i = \frac{V}{R} \text{ Electric Resistance}$$

$$\tan \theta = \frac{V}{i} = \text{Resistance}$$

As $T \uparrow \Rightarrow R \uparrow \Rightarrow \text{slope} \uparrow$

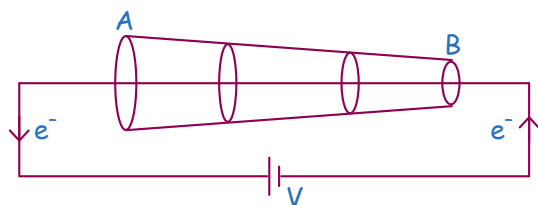
For a given material graph between ΔV & i is plotted at diff temp T_1 & T_2



$$(\text{Slope})_1 > (\text{Slope})_2$$

$$(T \uparrow, R \uparrow, \text{slope} \uparrow) \Rightarrow T_1 > T_2$$

Q. As we move from right to left A to B analyse the question.



Sol. A to B $\Rightarrow$ Area of cross section decrease

$$i = V_d e n A$$

1. Current $\rightarrow$ same

2. Current density $= \frac{i}{A} = J \uparrow$ (increase)

3. Electric field $= E \uparrow$ (increase bcz $\vec{J} = \sigma \vec{E}$)

4. Drift velocity $V_d \uparrow$ (increase bcz $V_d = \frac{i}{enA}$)

TEMP DEPENDENCY OF ρ & R

$$\star \rho = \rho_0(1 + \alpha \Delta T)$$

$$\star R = R_0(1 + \alpha \Delta T)$$

$$\star R = R_0[1 + \alpha(T - T_i)]$$

$\star R_0$ is the resistance at Temp T_i

$\star T_i = 0^\circ\text{C} \Rightarrow R_0$ is resistance at 0°C

$\star$ For metal/conductor $\Rightarrow \alpha > 0$ $T \uparrow, R \uparrow$

$\star$ For semi conductor $\Rightarrow \alpha < 0$ $T \uparrow, R \downarrow$

Q. Value of resistance at 10°C is 50Ω and at 30°C is 60Ω Find resistance at 50°C

Sol. $R = R_0(1 + \alpha \Delta T)$

$$50 = R_0[(1 + \alpha(10 - 0))]$$

$$60 = R_0[(1 + \alpha(30 - 0))]$$

$$60 + 600\alpha = 50 + 1500\alpha$$

$$\alpha = \frac{1}{90}$$

$$50 = R_0(1 + 1/90 \times 10)$$

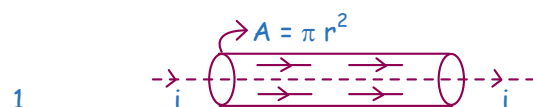
$$R_0 = 45$$

$$R_f = R_0(1 + \alpha \Delta T)$$

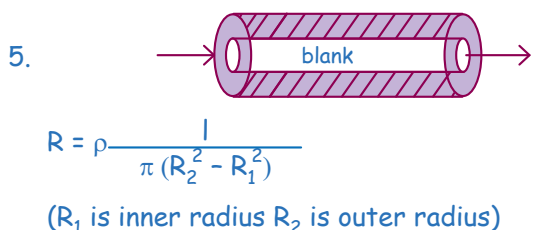
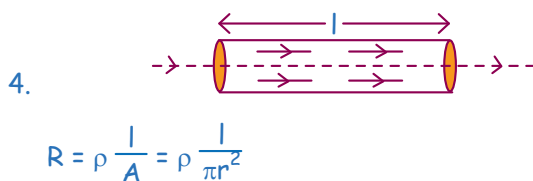
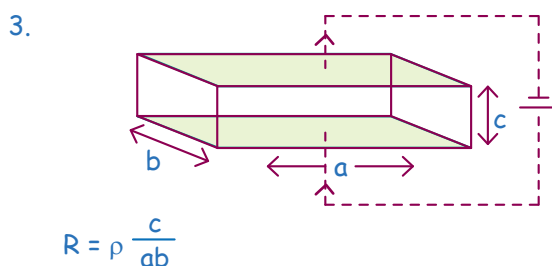
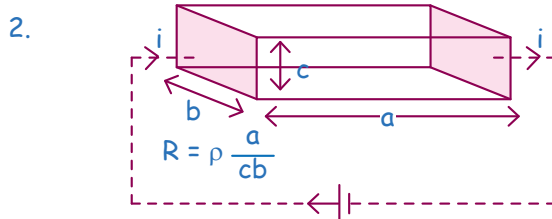
$$= 45(1 + 1/90 \times (50 - 0))$$

$$= 45(1 + 5/9) = \frac{45 \times 14}{9}$$

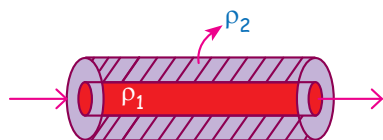
ab in following case mein R_{eq} ki ache se practice karlo (apply smartly $R = \rho l/A$)



$$\text{Solid Cylinder } R = \rho \frac{l}{\pi r^2}$$



6.

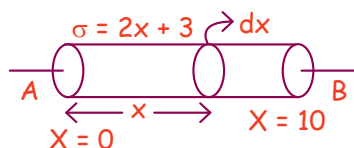


$$R_1 = R \text{ andar} = \rho_1 \frac{l}{\pi R_1^2}$$

$$R_2 = R \text{ baahar} = \rho_2 \frac{l}{\pi(R_2^2 - R_1^2)}$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

Q. R_{eq} between A & B



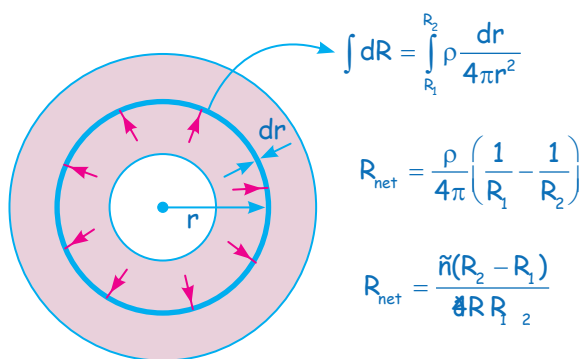
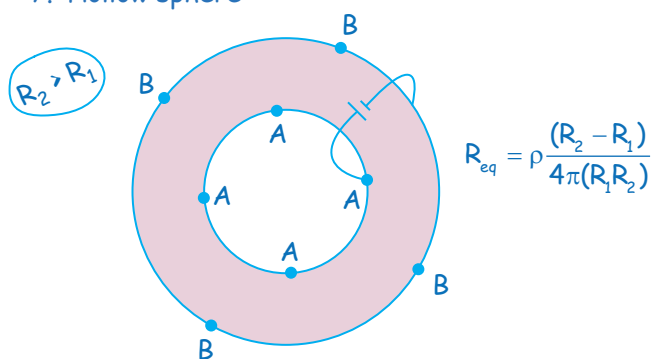
$$\text{Sol. } dR = \rho \frac{dx}{\pi r^2} = \frac{l}{\sigma} \frac{dx}{\pi r^2}$$

$$\int dR = \int \frac{1}{2x+3} \times \frac{dx}{\pi r^2}$$

$$R_{eq} = \frac{1}{\pi r^2} \int_0^{10} \frac{dx}{2x+3}$$

$$\text{Ans: } R = \frac{1}{\pi r^2 \times 2} \ln\left(\frac{23}{3}\right)$$

7. Hollow sphere

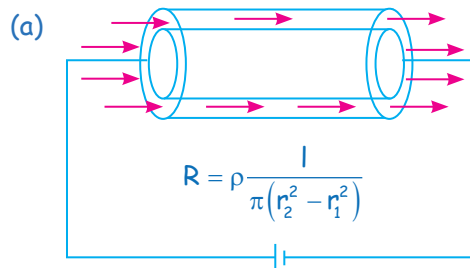
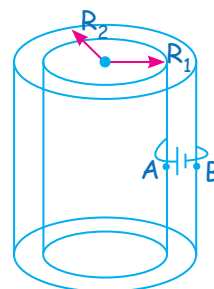


$$\int dR = \int_{R_1}^{R_2} \rho \frac{dr}{4\pi r^2}$$

$$R_{net} = \frac{\rho}{4\pi} \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$R_{net} = \frac{\rho(R_2 - R_1)}{4\pi R_1 R_2}$$

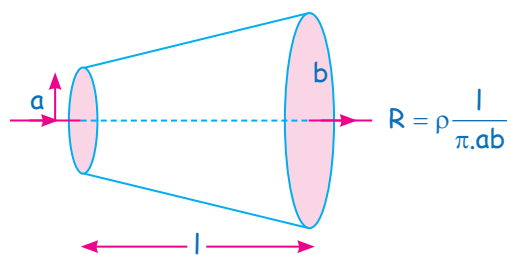
8. Hollow cylinder

(b) $R_2 > R_1$ 

$$\int dR = \int_{R_1}^{R_2} \rho \frac{dr}{2\pi r l}$$

$$R_{eq} = \frac{\rho}{2\pi l} \ln\left(\frac{R_2}{R_1}\right)$$

9.



Q. If length of resistance is increase by 5% Find % increase in resistance.

$$\text{Sol. } R = \rho \frac{l}{A} = \rho \frac{l^2}{A l} \quad (V \rightarrow \text{const})$$

$$R \propto l^2 \quad \frac{\Delta l}{l} \times 100 = 5$$

$$\frac{\Delta R}{R} = \frac{2\Delta l}{l}$$

$$\frac{\Delta R}{R} \times 100 = 2 \left(\frac{\Delta l \times 100}{l} \right)$$

$$= 2 \times 5 = 10\%$$

Q. A cylindrical wire is increased double its original length the % increase in the Resistance of wire

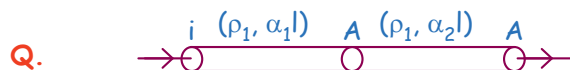
$$\text{Sol. } R = \frac{\rho l}{A} = \frac{\rho l^2}{A l} = \frac{2l^2}{\text{vol}^n}$$

$$R \propto l^2 \quad l \rightarrow \text{double}$$

$R \rightarrow 4$ times

$$\% \frac{\Delta R}{R} = \frac{R_f - R_i}{R} \times 100$$

Ans. 300%



Find the condition for which this combination R_{eq} is independent on the Temp

Sol. $R_i = R_1 + R_2 = \rho_1 \frac{l}{A} + \rho_2 \frac{l}{A}$

$$R_f = (R_1 \text{ naya}) + (R_2 \text{ naya}) = R_1(1 + \alpha_1 \Delta T) + R_2(1 + \alpha_2 \Delta T)$$

$$R_i = R_f$$

$$\rho_1 \frac{l}{A} + \rho_2 \frac{l}{A} = \rho_1 \frac{l}{A} (1 + \alpha_1 \Delta T) + \rho_2 \frac{l}{A} (1 + \alpha_2 \Delta T)$$

$$\text{Solve and get } \rho_1 \alpha_1 + \rho_2 \alpha_2 = 0$$

(Ab yah mat sochnaa aisaa kaise huaa iskaa matlab hai koī ek α negative hai)

CONVERSION OF GALVANOMETER INTO AMMETER/VOLTMETER.



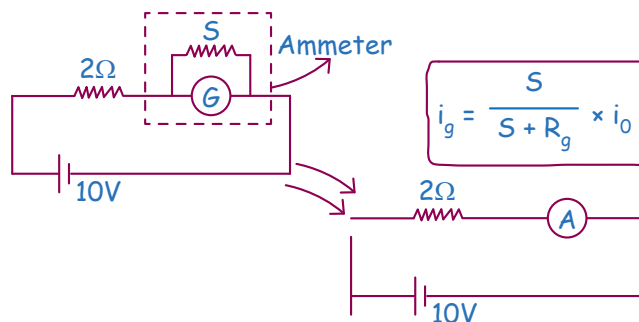
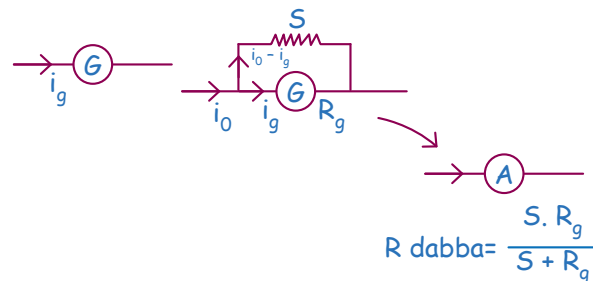
Teenon dikhne mein ek jaise hee hote hain bas labelling kaa antar hai yakeen nahin aataa upar ke image ko theek se dekhnaa.



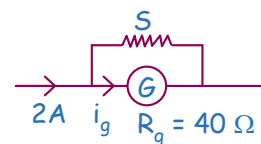
Bas itnaa yaad rakho galvanometer ke parallel mein chhotaa saa resistance lagaane par yah ammeter ban jaataa hai aur galvaometer ke baaju mein bahut badaa resistance lagaane par ye voltmeter ban jaataa hai.

Ideal ammeter kaa resistance zero or ideal voltmeter kaa resistance infinity hotaa hai.

Conversion of Galvanometer into Ammeter



Q. A Galvanometer of resistance 40Ω can read max current of 50mA . Find the resistance require to that it converted into ammeter which can measure the current upto 2A .



Sol. $i_g = \frac{S}{S + R_g} \times i_{\text{total}}$

$$50 \times 10^{-3} = \frac{S}{S + 40} \times 2$$

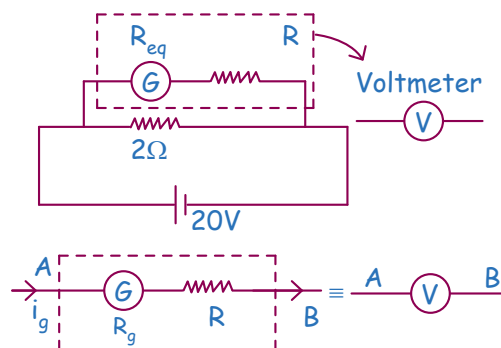
$$50(S + 40) = 2000S$$

$$50S + 2000 = 2000S$$

$$2000 = 1950S$$

$$S = \frac{2000}{1950}$$

Conversion of Galvanometer into Voltmeter

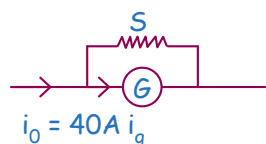


$$V_{AB} = i_g(R + R_g) \text{ (ideally V Resist infinity)}$$

Q. A galvanometer has coil of resistance 40Ω showing full scale deflection of 80mA what resistance should be added and how so that

1. It become Ammeter of range 40A

Sol.



$$i = \frac{S}{S+g} \times i_0$$

$$80 \times 10^{-3} = \frac{S}{S+40} \times 40$$

$$80S + 3200 = 40000 'S'$$

$$S = \frac{3200}{39920}$$

2. It become voltmeter of range 40 volt



$$V_{AB} = i_g(R_g + R)$$

$$40 = 80 \times 10^{-3}(40 + R)$$

$$R = 460\Omega$$

Now few important questions practice them sincerely

Q. In following case find r_{eq} & ρ_{eq} if rods A & B are connected in series of length $3l_0$.

Rod A: $\rho_0, l_0, 3\alpha, A$

Rod B $\Rightarrow 2\rho_0, 2l_0, 2\alpha, A$

Sol.

$$R_1 = \rho_0(1 + 3\alpha\Delta T) \frac{l_0}{A}$$

at only temp

$$R_2 = 2\rho_0(1 + 2\alpha\Delta T) \frac{2l_0}{A}$$

$$R_{eq} = R_1 + R_2 = \rho_0 \frac{l_0}{A}(1 + 3\alpha\Delta T) + 4\rho_0 \frac{l_0}{A}(1 + 2\alpha\Delta T)$$

$$\rho_{eq}(1 + \alpha_{eq}\Delta T) \frac{3l_0}{A} = R_{eq} = \rho_0 \frac{l_0}{A}(1 + 3\alpha\Delta T) + 4\rho_0 \frac{l_0}{A}(1 + 2\alpha\Delta T)$$

$$3\rho_{eq}(1 + \alpha_{eq}\Delta T) = \rho_0 + 3\alpha\rho_0\Delta T + 4\rho_0 + 8\alpha\rho_0\Delta T$$

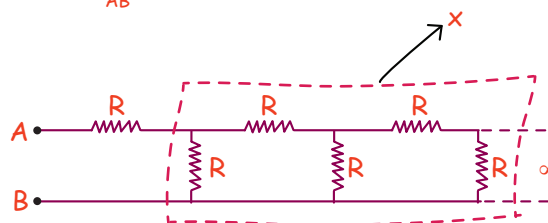
$$3\rho_{eq} + (3\rho_{eq}\alpha_{eq})\Delta T = 5\rho_0 + 11\alpha\rho_0\Delta T$$

$$3\rho_{eq} = 5\rho_0 \quad \rho_{eq} = 5/3\rho_0$$

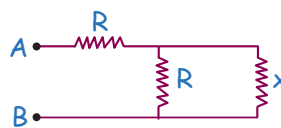
$$3\rho_{eq}\alpha_{eq} = 11\alpha\rho_0 \quad 5\rho_0\alpha_{eq} = 11\alpha\rho_0 \quad \alpha_{eq} = \frac{11}{5}\alpha$$

* ∞ ladder question

Q. Find R_{AB} ?



Sol.



Circuit ko shuru mein kitna mittu du ki uski sakal surat waise hi rahaae

$$R_{AB} = x$$

$$R_{AB} = x = R + \frac{R \times}{R + x}$$

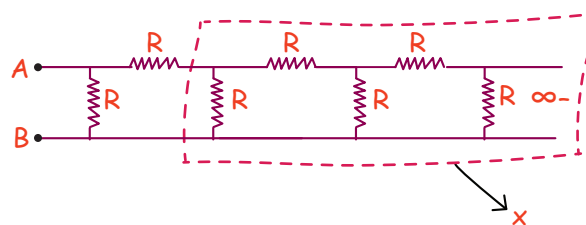
$$x = \frac{R(R + x) + Rx}{R + x}$$

$$xR + x^2 = R^2 + Rx + Rx$$

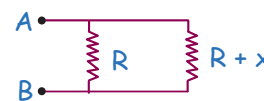
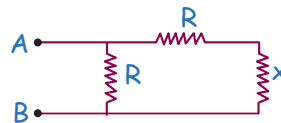
$$x^2 - Rx - R^2 = 0$$

$$x = \text{solve \& get, } x = \frac{R + \sqrt{5}R}{2}$$

Q. $R_{AB} = ?$



Sol.



$$R_{AB} = \frac{R(R + x)}{R + R + x} = x = \frac{R^2 + Rx}{2R + x} = x$$

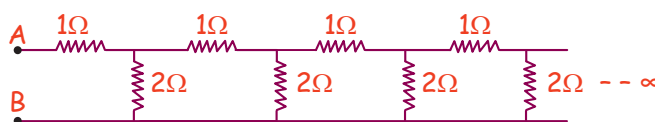
$$R^2 + Rx = 2Rx + x^2$$

$$x^2 - Rx - R^2 = 0$$

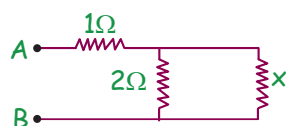
$$x = \frac{-R + \sqrt{R^2 + 4R^2}}{2}$$

$$x = R \left(\frac{\sqrt{5} - 1}{2} \right)$$

Q. $R_{AB} = ?$



Sol.



$$R_{AB} = x = 1 + \frac{2x}{x+2}$$

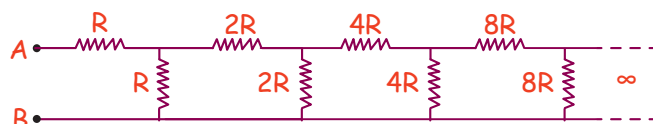
$$x - 1 = \frac{2x}{x+2}$$

$$(x - 1)(x + 2) = 2x$$

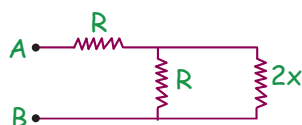
$$x^2 + 2x - x - 2 = 2x$$

$$x = \frac{1 - \sqrt{1+8}}{2} = 2$$

Q. Find $R_{AB} = ?$



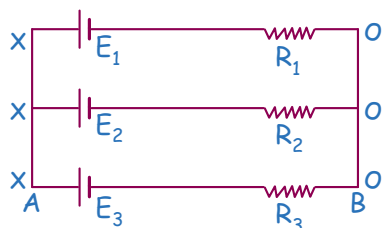
Sol.



$$R_{AB} = x = \frac{R \cdot 2x}{R + 2x} + R$$

Solve & get

Grouping of Cell



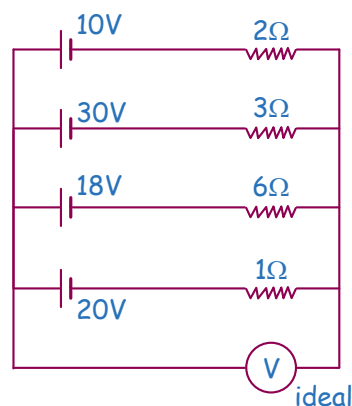
#SKC

$$x = V_{AB} = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2} + \frac{E_3}{r_3}}{\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}}$$

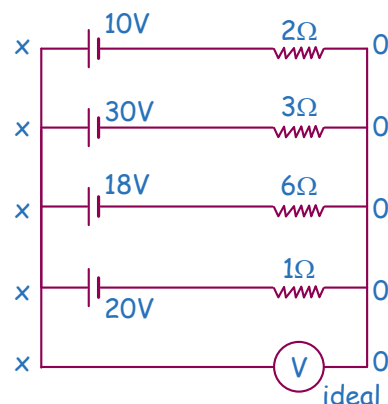
agar is tarh ke question mein direct x pucho toh bindas SKC lagao.



Q. Find reading of ideal voltmeter.



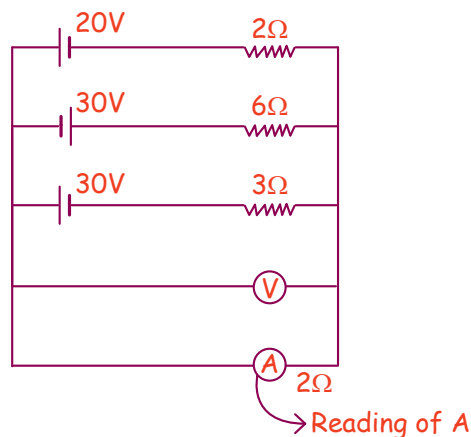
Sol.



$$x = \frac{10/2 + 30/3 + 18/6 + 20/1}{1/2 + 1/3 + 1/6 + 1/1} = \frac{5 + 10 + 3 + 20}{2}$$

$$x = V_{AB} = 19$$

SSSQ. Find volt meter and ammeter reading

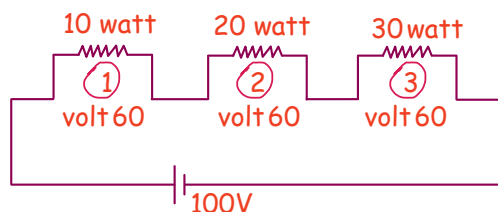


$$\text{Sol. } x = \frac{20/2 - 30/6 + 30/3 + 0/2}{1/2 + 1/6 + 1/3 + 1/2} = 10$$

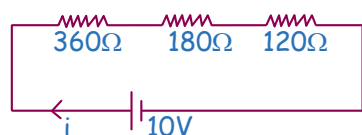
$$= 10 \text{ (V) (Voltmeter Reading)}$$

$$i = \frac{x - 0}{2} = \frac{10 - 0}{2} = 5 = \text{Ammeter reading}$$

Q. Compare brighter of bulbs.



Sol. $R = \frac{V^2}{P} = \frac{60 \times 60}{10} = 360$



$B_1 > B_2 > B_3$

$i = \frac{10}{360 + 180 + 120}$

Q. Find the total average momentum of electrons in a straight wire of length $l = 1000$ m carrying current $I = 704$ A.

Sol. Let n be no. of electrons per unit volume.

No. of electrons in length l

$N = nSl$ (S is cross-sectional area)

Momentum of one electron $= mv_d$

Total momentum $P = (nSl)mv_d$

As $v_d = \frac{I}{neS}$

$P = (nSl)m \frac{I}{(neS)} = \frac{mI}{e}$

On substituting numerical values, we get

$P = 4\mu\text{Ns}$

Q. The temperature coefficient of resistivity α is given by $\alpha = \left(\frac{1}{\rho}\right) \frac{d\rho}{dT}$, where ρ is the resistivity at temperature T . Assume that α is not constant and follows the relation $\alpha = -\frac{a}{T}$, where T is the absolute temperature and a is a constant. Show that the resistivity ρ is given by $\rho = \frac{b}{T^a}$, where b is another constant.

Sol. $\alpha = \frac{1}{\rho} \frac{d\rho}{dT} \Rightarrow \frac{d\rho}{\rho} = \alpha dT = -a \frac{dT}{T}$

Let $\rho = \rho_0$ at $T = T_0$, then

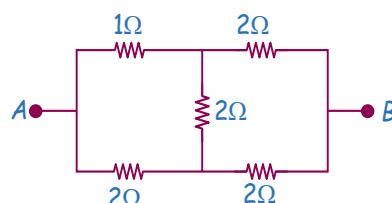
$\int_{\rho_0}^{\rho} \frac{d\rho}{\rho} = -a \int_{T_0}^T \frac{dT}{T}$

$\Rightarrow \log_e \left(\frac{\rho}{\rho_0} \right) = -a \log_e \left(\frac{T}{T_0} \right) = \log_e \left(\frac{T_0}{T} \right)^a$

$\Rightarrow \rho = (\rho_0 T_0^a) \frac{1}{T^a} = \frac{b}{T^a}$

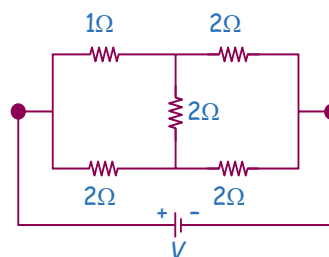
Here, $b = \rho_0 T_0^a$

Q. Find the equivalent resistance across terminals A and B.

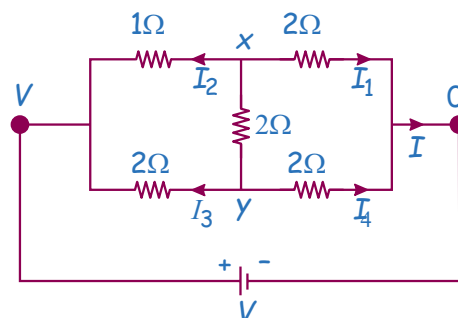


Sol. This is a case of unbalanced Wheatstone bridge.

Step 1: Connect a battery across the terminal.



Step 2: Mark the voltages of nodes.



Step 3: Calculate $R_{eq} = \frac{V}{I}$

Apply KCL at node 'x'

$\Rightarrow \frac{x-V}{1} + \frac{x-0}{2} + \frac{x-y}{2} = 0$

$\Rightarrow 2x - 2V + x + x - y = 0 \Rightarrow 4x - 2V = y \dots (i)$

Apply KCL at node 'y'

$\frac{y-x}{2} + \frac{y-V}{2} + \frac{y}{2} = 0 \Rightarrow 3y - V = x \dots (ii)$

Solve equations (i) and (ii),

$x = \frac{7V}{11}, y = \frac{6V}{11}$

Now calculate: $I_1 = \frac{x-0}{2} = \frac{7V}{22}$ and $I_2 = \frac{y}{2} = \frac{6V}{22}$

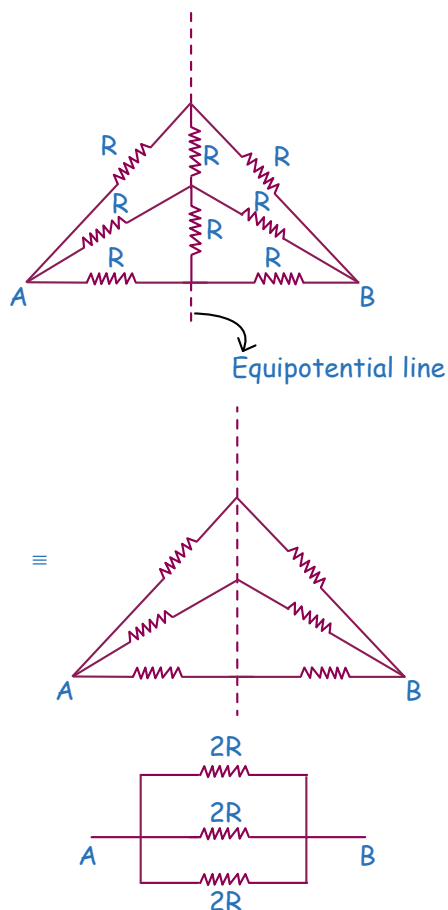
So, equivalent resistance

$$R_{eq} = \frac{V}{I} = \frac{V}{I_1 + I_2} = \frac{V}{\frac{7V}{22} + \frac{6V}{22}} = \frac{V}{\frac{13V}{22}} \Rightarrow R_{eq} = \frac{22}{13} \Omega$$

SYMMETRY (Not much important for jee mains)

- Perpendicular Bisector symm
- Folding symm
- input output symm

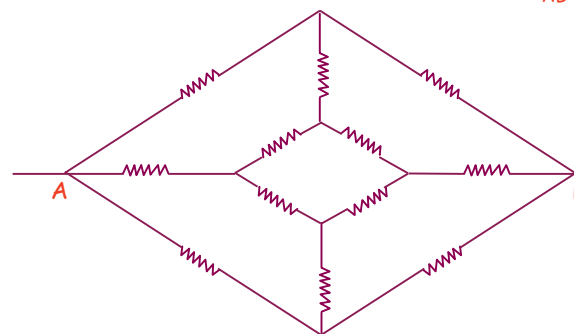
Perpendicular Bisector symmetry example



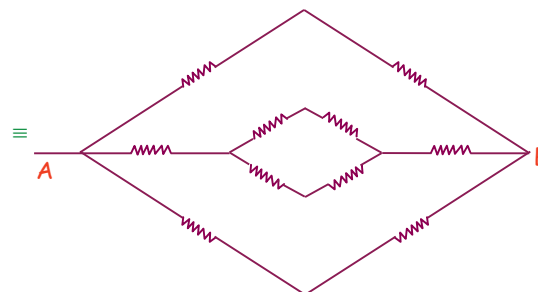
#SKC

A ko B se connect karo aur AB ke $\perp^r$ Bisector lo uske aaj baju agar circuit mirror image hai toh bisector line ke resistance uda do

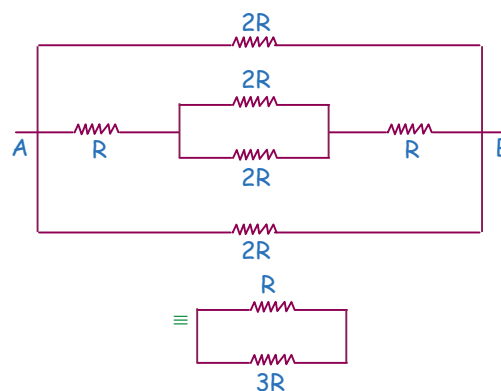
Q. All resistors have same value 'R'. Find R_{AB}



Sol.



All resistance have same 'R'



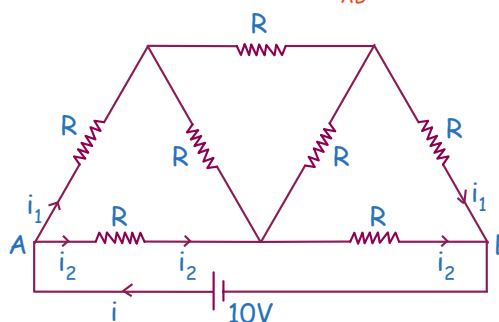
$$\frac{3R}{4} = R_{AB}$$

✦ Input output symmetry

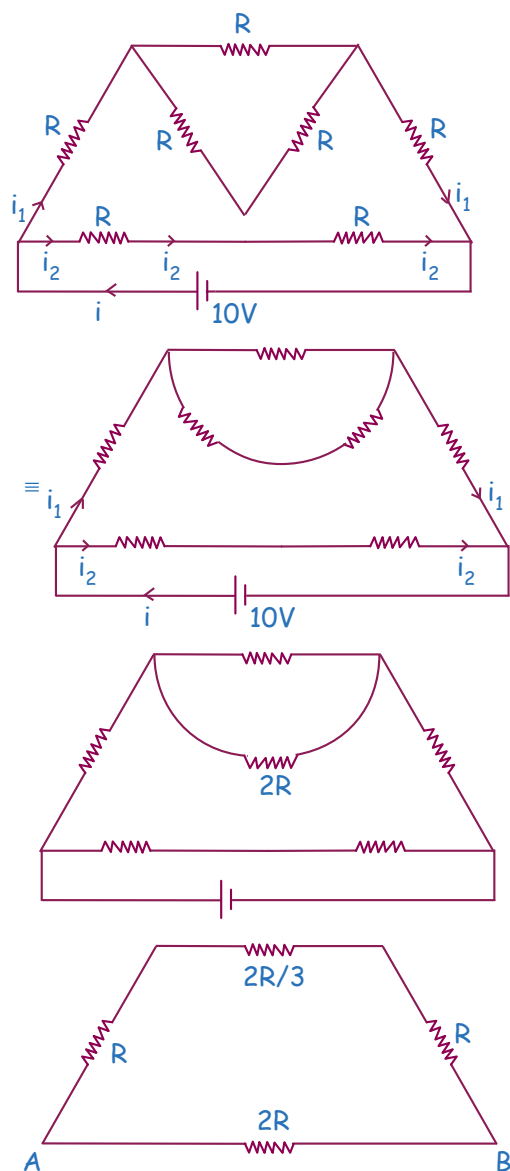
If current has similar path in entering side to exit & vice versa then circuit is said to be input output symmetry.

Under such condition for entry side & exit side are same

Q. Each resistance = R, Find R_{AB}



Sol.

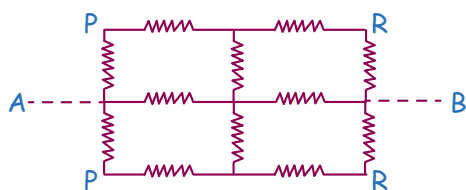


Folding Symmetry

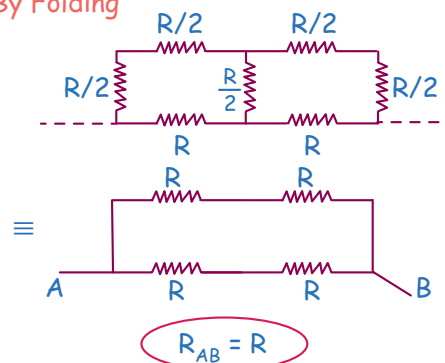
#SKC

agar AB line ke upar niche circuit ek ... jaisa ho to neeche wale ko upar wale par fold kar do matlab upar ke saare resistance aadhe kar do.

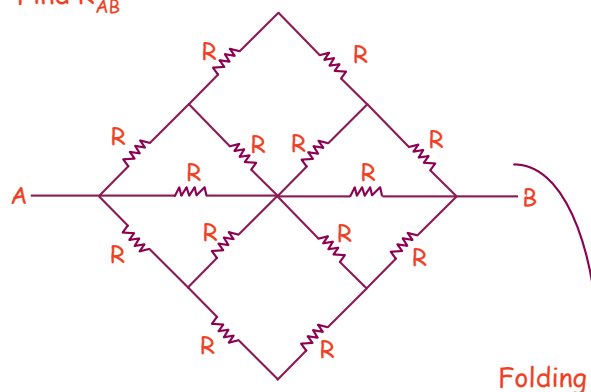
Q. Each resistance = R , Find R_{AB}



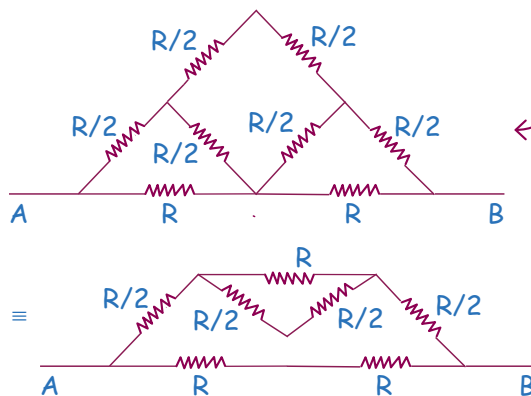
By Folding



Q. Find R_{AB}

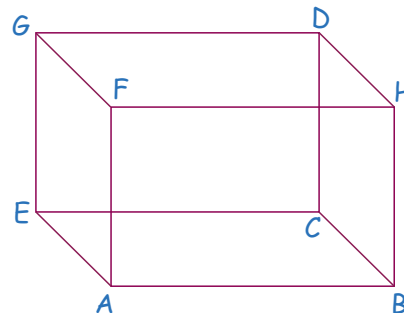


Sol.



Now you can solve.

✦ Cube each of length l , Resistance R (each)



$$\frac{7R}{12} \quad \frac{3R}{4} \quad \frac{5R}{6}$$

R_{AB} R_{AC} R_{AD}

pass pass thoda dur dur dur

- ★ If two appliances with rating (W_1, v) & (W_2, v) are connected to V .

1. In Parallel



$$\text{Total power dissipated} = \omega_1 + \omega_2$$

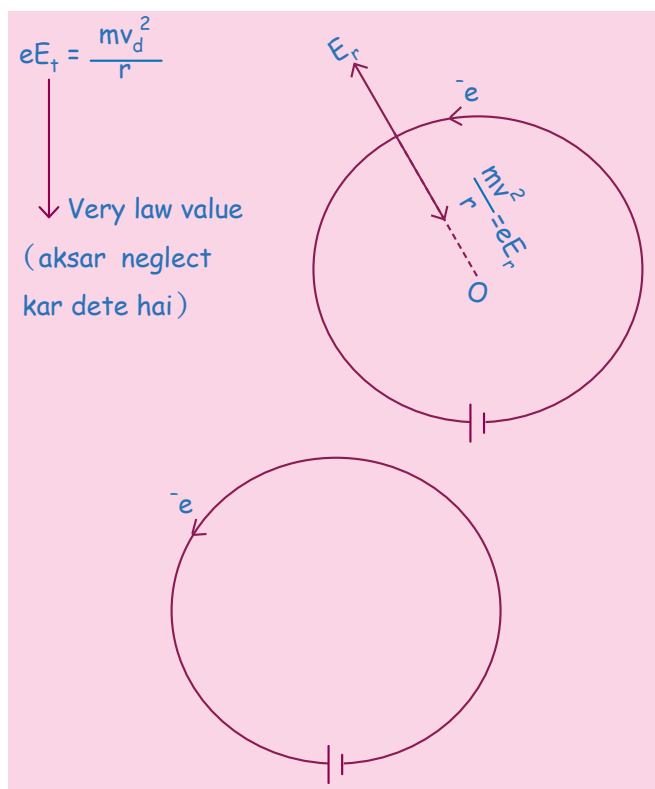
2. In series



$$\text{Total power dissipated} = \omega_{\text{net}}$$

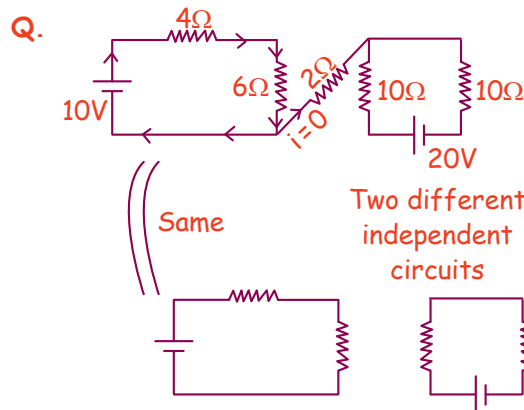
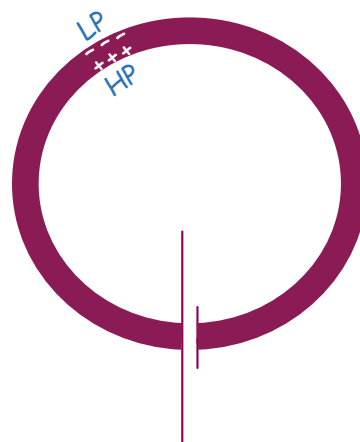
$$\frac{1}{\omega_{\text{net}}} = \frac{1}{\omega_1} + \frac{1}{\omega_2}$$

Q. A wire is in a circular shape connected to a battery as shown in figure. Find value of radial electric field.

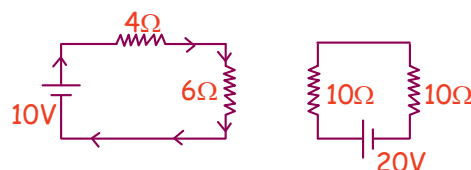


#SKC

Yah dhyaan se dekhnaa
Current kiski wajah se aayaa? e^- ki
wajah ya +ve charge ke motion
ki wajah se



Now solve and get

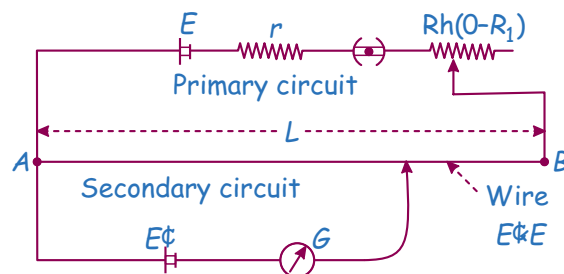


POTENTIOMETER (Not in Jee Mains 2025)

Working Principle of Potentiometer

Any unknown potential difference is balanced on a known potential difference which is uniformly distributed over entire length of potentiometer wire. This process is named as zero deflection or null deflection method.

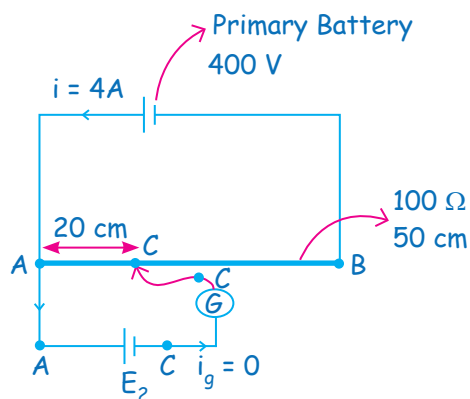
Circuit of Potentiometer



Primary circuit contains constant source of voltage and a rheostat (a resistance box).

Secondary, unknown or galvanometer circuit contains components with unknown parameters.

Q. Find emf of the battery if galvanometer show null deflection at C.



$$V_{AB} = 400 \text{ V}$$

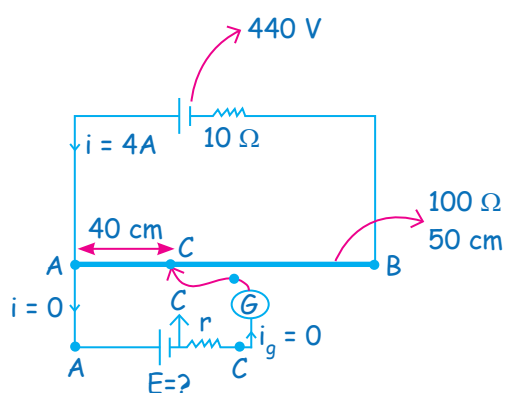
$$50 \text{ cm} \longrightarrow 400 \text{ V}$$

$$1 \text{ cm} \longrightarrow \frac{400}{50}$$

$$20 \text{ cm} \longrightarrow \frac{400}{50} \times 20$$

$$= 160 = V_{AC} = E_2$$

Q. Find emf of the battery if galvanometer show null deflection at C.

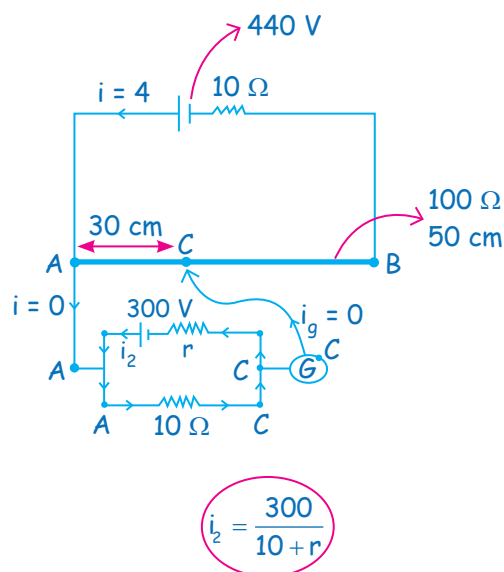


$$V_{AB} = 440 \text{ V}$$

$$50 \text{ cm} \longrightarrow 440 \text{ V}$$

$$40 \text{ cm} \longrightarrow \frac{440}{50} \times 40 = 352 \text{ V}$$

Q. Find internal resistance of the battery if galvanometer show null deflection at C.



$$i_2 = \frac{300}{10 + r}$$

$$V_{AB} = 300 \text{ V}$$

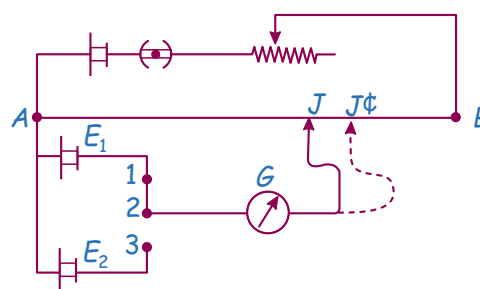
$$V_{AC} = \frac{300}{50} \times 30 = 180$$

$$V_{AC} = \frac{300}{10 + r} \times 10$$

$$180 = \frac{3000}{10 + r}$$

$$r = 2.5 \Omega$$

☆ Comparison of emf of two cells



Plug only in (1- 2): Balance length $AJ = \ell_1$

Plug only in (2 - 3): Balance length $AJ = \ell_2$

$$E_1 = x \ell_1 \text{ and } E_2 = x \ell_2$$

$$\Rightarrow \frac{E_1}{E_2} = \frac{\ell_1}{\ell_2}$$

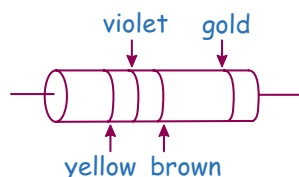
COLOUR CODE FOR CARBON RESISTORS (Removed from Mains 2025 and in Advance also)



	Strip A	Strip B	Strip C	Strip D
Colour	(digit 1)	(digit 2)	(Multiplier)	(Tolerance)
Black	0	0	10^0	
Brown	1	1	10^1	
Red	2	2	10^2	
Orange	3	3	10^3	
Yellow	4	4	10^4	
Green	5	5	10^5	
Blue	6	6	10^6	
Violet	7	7	10^7	
Grey	8	8	10^8	
White	9	9	10^9	
Gold	-	-	10^{-1}	$\pm 5\%$
Silver	-	-	10^{-2}	$\pm 10\%$
No Colour	-	-	-	$\pm 20\%$

○ Aid to memory **BBROY** of **G**reat **B**ritain does a **V**ery **G**ood **W**ork.

Q. What is the resistance of the following resistor?



Sol. Number for yellow is 4. Number for violet is 7.

Brown colour gives multiplier 10^1 , Gold gives a tolerance of $\pm 5\%$

So, resistance of resistor is

$$47 \times 10^1 \Omega \pm 5\% = (470 \pm 5\%) \Omega$$